

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\LP Nitrogen Buffer Vessel (V8601)Rev 7%.pvdb

Date of Analysis : Sep 16, 2026 2:48pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 6000.000 mm. ID
 Vessel Design Length, Tangent to Tangent 26000.00 mm.
 Distance of Bottom Tangent above Grade 3800.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 3500.00 mm.
 Internal Design Temperature 82 °C
 Internal Design Pressure 13.500 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 14 bars
 Shop Test Pressure 18.425 bars
 Field Test Pressure 17.550 bars
 Required Minimum Design Metal Temperature -1.0 °C
 Warmest Computed Minimum Design Metal Temperature 4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
SKIRT-1	...	...	...	1.0000	N/A	No
SKIRT-2	...	...	...	1.0000	N/A	No
BOTTOM HEAD	13.564	0.00	14.000	1.6000	No	No
SHELL-1	13.500	0.00	14.000	1.6000	N/A	No
SHELL-2	13.500	0.00	14.000	1.6000	N/A	No
SHELL-3	13.500	0.00	14.000	1.6000	N/A	No
SHELL-4	13.500	0.00	14.000	1.6000	N/A	No
SHELL-5	13.500	0.00	14.000	1.6000	N/A	No
SHELL-6	13.500	0.00	14.000	1.6000	N/A	No
SHELL-7	13.500	0.00	14.000	1.6000	N/A	No
SHELL-8	13.500	0.00	14.000	1.6000	N/A	No
SHELL-9	13.500	0.00	14.000	1.6000	N/A	No
TOP HEAD	13.500	0.00	14.000	1.6000	No	No



Liquid Level: 662.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.989

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1500.0	2000.0	22.0	22.0	...	...	0.70	0.70
Skirt	0.0	1500.0	22.0	22.0	...	...	0.70	0.50
Ellipse	50.0	50.0	38.0	32.0	31.1	...	1.00	1.00
Cylinder	3050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	6050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	9050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	12050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	15050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	18050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	21050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	24050.0	3000.0	32.0	32.0	31.1	...	1.00	1.00
Cylinder	25950.0	1900.0	32.0	32.0	31.1	...	1.00	1.00
Ellipse	26000.0	50.0	38.0	32.0	31.0	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	1.138	n/a	Failed
Trunnion	WRC-107/537	1.139	n/a	Failed
Trunnion	WRC-107/537	1.140	n/a	Failed
Trunnion	WRC-107/537	1.141	n/a	Failed
Trunnion	WRC-107/537	1.142	n/a	Failed
Trunnion	WRC-107/537	1.142	n/a	Failed
Trunnion	WRC-107/537	1.141	n/a	Failed
Trunnion	WRC-107/537	1.141	n/a	Failed
Trunnion	WRC-107/537	1.140	n/a	Failed
Trunnion	WRC-107/537	1.138	n/a	Failed
Trunnion	WRC-107/537	1.137	n/a	Failed
Trunnion	WRC-107/537	1.135	n/a	Failed
Trunnion	WRC-107/537	1.132	n/a	Failed
Trunnion	WRC-107/537	1.130	n/a	Failed
Trunnion	WRC-107/537	1.127	n/a	Failed
Trunnion	WRC-107/537	1.123	n/a	Failed
Trunnion	WRC-107/537	1.119	n/a	Failed
Trunnion	WRC-107/537	1.115	n/a	Failed
Trunnion	WRC-107/537	1.111	n/a	Failed
Trunnion	WRC-107/537	1.106	n/a	Failed
Trunnion	WRC-107/537	1.101	n/a	Failed
Trunnion	WRC-107/537	1.095	n/a	Failed
Trunnion	WRC-107/537	1.090	n/a	Failed
Trunnion	WRC-107/537	1.084	n/a	Failed
Trunnion	WRC-107/537	1.077	n/a	Failed
Trunnion	WRC-107/537	1.070	n/a	Failed
Trunnion	WRC-107/537	1.063	n/a	Failed
Trunnion	WRC-107/537	1.056	n/a	Failed
Trunnion	WRC-107/537	1.048	n/a	Failed
Trunnion	WRC-107/537	1.040	n/a	Failed
Trunnion	WRC-107/537	1.031	n/a	Failed
Trunnion	WRC-107/537	1.023	n/a	Failed
Trunnion	WRC-107/537	1.013	n/a	Failed
Trunnion	WRC-107/537	1.004	n/a	Failed
Trunnion	WRC-107/537	0.994	n/a	Passed
Trunnion	WRC-107/537	0.984	n/a	Passed
Trunnion	WRC-107/537	0.974	n/a	Passed
Trunnion	WRC-107/537	0.963	n/a	Passed
Trunnion	WRC-107/537	0.953	n/a	Passed
Trunnion	WRC-107/537	0.941	n/a	Passed
Trunnion	WRC-107/537	0.930	n/a	Passed
Trunnion	WRC-107/537	0.918	n/a	Passed



Trunnion	WRC-107/537	0.906	n/a	Passed
Trunnion	WRC-107/537	0.894	n/a	Passed
Trunnion	WRC-107/537	0.881	n/a	Passed
Trunnion	WRC-107/537	0.868	n/a	Passed
Trunnion	WRC-107/537	0.855	n/a	Passed
Trunnion	WRC-107/537	0.842	n/a	Passed
Trunnion	WRC-107/537	0.828	n/a	Passed
Trunnion	WRC-107/537	0.814	n/a	Passed
Trunnion	WRC-107/537	0.800	n/a	Passed
Trunnion	WRC-107/537	0.785	n/a	Passed
Trunnion	WRC-107/537	0.771	n/a	Passed
Trunnion	WRC-107/537	0.756	n/a	Passed
Trunnion	WRC-107/537	0.740	n/a	Passed
Trunnion	WRC-107/537	0.725	n/a	Passed
Trunnion	WRC-107/537	0.709	n/a	Passed
Trunnion	WRC-107/537	0.694	n/a	Passed
Trunnion	WRC-107/537	0.677	n/a	Passed
Trunnion	WRC-107/537	0.661	n/a	Passed
Trunnion	WRC-107/537	0.645	n/a	Passed
Trunnion	WRC-107/537	0.628	n/a	Passed
Trunnion	WRC-107/537	0.611	n/a	Passed
Trunnion	WRC-107/537	0.594	n/a	Passed
Trunnion	WRC-107/537	0.577	n/a	Passed
Trunnion	WRC-107/537	0.559	n/a	Passed
Trunnion	WRC-107/537	0.562	n/a	Passed
Trunnion	WRC-107/537	0.569	n/a	Passed
Trunnion	WRC-107/537	0.575	n/a	Passed
Trunnion	WRC-107/537	0.582	n/a	Passed
Trunnion	WRC-107/537	0.589	n/a	Passed
Trunnion	WRC-107/537	0.596	n/a	Passed
Trunnion	WRC-107/537	0.604	n/a	Passed
Trunnion	WRC-107/537	0.611	n/a	Passed
Trunnion	WRC-107/537	0.619	n/a	Passed
Trunnion	WRC-107/537	0.627	n/a	Passed
Trunnion	WRC-107/537	0.636	n/a	Passed
Trunnion	WRC-107/537	0.645	n/a	Passed
Trunnion	WRC-107/537	0.655	n/a	Passed
Trunnion	WRC-107/537	0.665	n/a	Passed
Trunnion	WRC-107/537	0.676	n/a	Passed
Trunnion	WRC-107/537	0.688	n/a	Passed
Trunnion	WRC-107/537	0.702	n/a	Passed
Trunnion	WRC-107/537	0.717	n/a	Passed
Trunnion	WRC-107/537	0.735	n/a	Passed
Trunnion	WRC-107/537	0.755	n/a	Passed
Trunnion	WRC-107/537	0.779	n/a	Passed
Trunnion	WRC-107/537	0.807	n/a	Passed
Trunnion	WRC-107/537	0.843	n/a	Passed
Trunnion	WRC-107/537	0.888	n/a	Passed
Trunnion	WRC-107/537	0.947	n/a	Passed
Trunnion	WRC-107/537	1.142	n/a	Failed
N1-10"	WRC-297	0.964	Shell	Passed
N3-8"	WRC-107/537	0.869	n/a	Passed
N4-6"	WRC-107/537	0.869	n/a	Passed
N5-2"	WRC-107/537	0.869	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	13023.5	38293.9	214442	846609
20	30	2750	12418.2	38185	189000	770129
30	40	3525	11964.2	38040.7	170713	712958
40	50	5050	11588.9	36673	170124	711090
50	60	8050	10680.5	35900.3	136719	602228
60	70	11050	9398.45	34286	106600	496947
70	80	14050	8075.07	32212.4	80389.6	397198
80	90	17050	6727.65	29598.4	58185.1	304480

90	100	20050	5819.26	26997.8	39364.4	219584
100	110	23050	4506.13	23267.5	23876	144184
110	120	26050	3163.21	18979	12371.7	80813
120	130	28500	2254.82	14996.3	4244.51	29849
130	140	29475	1295.14	9611.5	871.984	6471.18

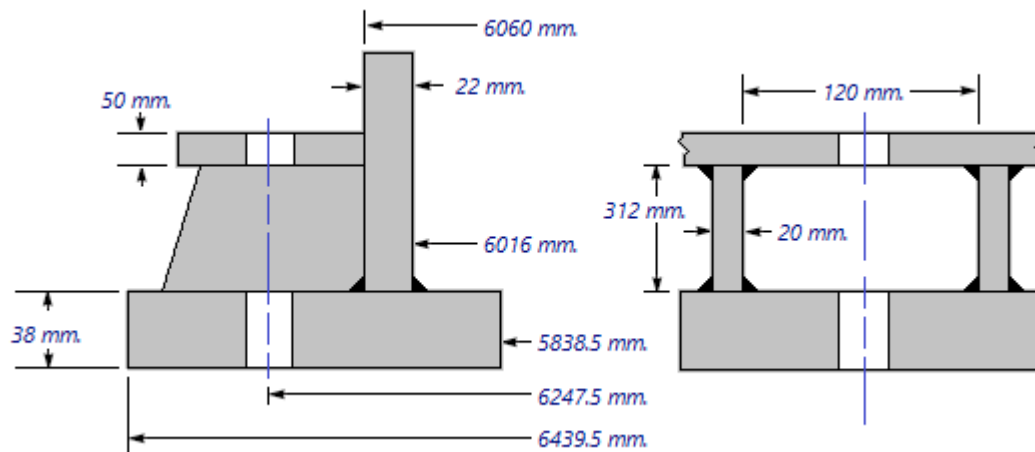
Abs Max of the all of the Stress Ratio's : 0.4779

Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering	38.0000	mm.
Inside Diameter of Basering	5838.5000	mm.
Outside Diameter of Basering	6439.5000	mm.
Nominal Diameter of Bolts	64.0000	mm.
Diameter of Bolt Circle	6247.5000	mm.
Number of Bolts	24	

Thickness of Gusset Plates	20.0000	mm.
Average Width of Gusset Plates	182.0000	mm.
Height of Gussets	312.0000	mm.
Distance between Gussets	120.0000	mm.
Thickness of Top Plate or Ring	50.0000	mm.
Radial Width of the Top Plate	192.0000	mm.

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	13023.	Kgf
Total Earthquake Shear on Support	38294.	Kgf
Wind Moment on Support	232537.	Kg-m.
Earthquake Moment on Support	864704.	Kg-m.
Maximum Initial Bolt Load	13211.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	179141.5	kg.
Shop Test - Fabricated + water (full)	977716.5	kg.
Shipping - Fab. + rem. intls.+ shipping app.	179141.5	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	226654.9	kg.
Empty - Fab. + intls. + details + wghts.	226654.9	kg.
Operating - Empty + operating liquid (no ca)	244221.1	kg.
Field Test - Empty weight + test medium (full)	1017994.8	kg.





Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-2"	14	...	...	OK	6.40	OK	Passed
N6-2"	14	...	...	OK	6.86	OK	Passed
K1B-3"	14	...	...	OK	7.33	OK	Passed
M1-24"	14	...	...	...		OK	Passed
M2-24"	14	...	...	...		OK	Passed
K1A-3"	14	...	...	OK	7.33	OK	Passed
N1-10"	14	...	...	OK	9.70	OK	Passed
K2-2"	14	...	...	OK	6.86	OK	Passed
N3-8"	14	...	...	OK	8.76	OK	Passed
N4-6"	14	...	...	OK	7.82	OK	Passed
N5-2"	14	...	...	OK	6.40	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 14 Nozzle : N5-2"
 Minimum MAWP Shells/Flanges : 14 Element : SHELL-9
 Minimum MAPnc Shells/Flanges : 15 Element : SHELL-9

Computed Vessel M.A.W.P. : 14 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N2-2"	0.000	0.000	145.128
40	N6-2"	1550.000	270.000	165.448
40	K1B-3"	1550.000	90.000	187.800
50	M1-24"	4550.000	110.000	1225.600
50	M2-24"	4550.000	250.000	1225.600
110	K1A-3"	22550.000	90.000	187.800
120	N1-10"	25000.002	320.000	501.700
120	K2-2"	25000.002	50.000	165.448
130	N3-8"	0.000	0.000	393.750
130	N4-6"	0.000	270.000	299.059
130	N5-2"	0.000	90.000	145.128

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	192.16	300
N6-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	219.78	900
K2-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	292.46	900
N5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	795.60	300
K1B-3"	3.000 in	Actual	LWN	5.000	25.400	...	...	292.67	900
K1A-3"	3.000 in	Actual	LWN	5.000	25.400	...	...	292.67	900
N4-6"	6.000 in	XS	WNF	6.625	10.973	350.00	32.00	603.51	150
N3-8"	8.000 in	XS	WNF	8.625	12.700	550.00	32.00	483.11	150
N1-10"	10.000 in	XS	WNF	10.750	12.700	450.00	32.00	295.11	150
M1-24"	24.000 in	Actual	WNF	25.575	20.000	1150.00	32.00	399.63	150
M2-24"	24.000 in	Actual	WNF	25.575	20.000	1150.00	32.00	399.63	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N2-2"	SA-105	38.000	10.000	...	...	...
N6-2"	SA-105	38.000	10.000	...	...	...
K2-2"	SA-105	38.000	10.000	...	...	...
N5-2"	SA-105	38.000	10.000	...	...	...
K1B-3"	SA-105	38.000	10.000	...	...	...
K1A-3"	SA-105	38.000	10.000	...	...	...
N4-6"	SA-106 B	32.000	10.000	30.000	32.000	...
N3-8"	SA-106 B	32.000	10.000	30.000	32.000	...
N1-10"	SA-106 B	32.000	10.000	30.000	32.000	...
M1-24"	SA-516 70	32.000	10.000	30.000	32.000	...
M2-24"	SA-516 70	32.000	10.000	30.000	32.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N2-2"	...	0.0	160.00	0.00	BOTTOM HEAD
N6-2"	1550.000	270.0	187.32	0.00	SHELL-1
K2-2"	25000.000	50.0	260.00	0.00	SHELL-9
N5-2"	...	90.0	750.00	0.00	TOP HEAD
K1B-3"	1550.000	90.0	260.00	0.00	SHELL-1
K1A-3"	22550.000	90.0	260.00	0.00	SHELL-8
N4-6"	...	270.0	550.00	0.00	TOP HEAD
N3-8"	...	0.0	450.00	0.00	TOP HEAD
N1-10"	25000.000	320.0	260.00	0.00	SHELL-9
M1-24"	4550.000	110.0	350.00	0.00	SHELL-2
M2-24"	4550.000	250.0	350.00	0.00	SHELL-2



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 38.0mm. THK X 6000.0mm. ID X 50.0mm.	SA-516 70
7	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
5	LADDER: 4000.0mm	...
1	LADDER: 4500.0mm	...
2	TRUNNIONS:	SA-516 70
2	TRUNNION PADS: 1050.0mm OD X 800.0mm ID X 32.0mm THK	
1	PLATFORM: 1000mm X 8500mm WIDE	
1	LADDER: 4100.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 6439.5mm. OD X 5838.5 ID X 38.0mm.	SA-516 70
24	BASERING BOLTS: 64.0mm. DIA	SA-193 B7
24	BASERING BOLT NUTS: 64.0mm. DIA	...
1	BASERING TOP RING: 6446.0mm. OD X 6062.0 ID X 50.0mm.	...
48	BASERING GUSSET PLATES: 182.0mm. X 182.0mm. X 312.0mm. X 20.0mm.	...
6	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2625.8mm. X 2000.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2573.3mm. X 2000.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 18328.1mm. X 2000.0mm.	SA-36
6	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2625.8mm. X 1500.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2573.3mm. X 1500.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 18328.1mm. X 1500.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 7487.3mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 7435.0mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 3926.9mm. X 3000.0mm.	SA-516 70
21	CYLINDRICAL SEGMENT 32.0mm. THK X 2617.9mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 2565.6mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 16231.3mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 32.0mm. THK X 575.9mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 16283.6mm. X 3000.0mm.	SA-516 70
12	CYLINDRICAL SEGMENT 32.0mm. THK X 3036.8mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 3089.2mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 15812.4mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 12199.6mm. X 3000.0mm.	SA-516 70



5	CYLINDRICAL SEGMENT 32.0mm. THK X 2617.9mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 16231.3mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 2565.6mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 575.9mm. X 1900.0mm.	SA-516 70
1	NAMEPLATE	...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BOTTOM HEAD	[10]	D	-29	-31		0.966	32.000	1.00	No
BOTTOM HEAD	[7]	D	-25	-36		0.811	38.000	1.00	No
SHELL-1	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-2	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-3	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-4	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-5	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-6	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-7	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-8	[8]	D	-29	-31		0.972	32.000	1.00	No
SHELL-9	[8]	D	-29	-31		0.972	32.000	1.00	No
TOP HEAD	[10]	D	-29	-31		0.966	32.000	1.00	No
TOP HEAD	[7]	D	-25	-36		0.811	38.000	1.00	No
N2-2"	[1]	A	7	-104		0.018	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-2"	[1]	A	21	-104		0.010	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B-3"	[1]	A	19	-104		0.016	25.400	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	B	6	4		0.972	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-35					
M2-24"	[1]	B	6	4		0.972	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-35					
K1A-3"	[1]	A	19	-104		0.016	25.400	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-10"	[1]	B	6	4		0.972	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-35					
K2-2"	[1]	A	21	-104		0.010	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-8"	[1]	B	6	-2		0.870	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-35					
N4-6"	[1]	B	6	-2		0.870	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-35					
N5-2"	[1]	A	7	-104		0.018	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					

Warmest MDMT: 21 4

Required Minimum Design Metal Temperature -1.0 °C
 Warmest Computed Minimum Design Metal Temperature 4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material



[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Warn.	10	20	Skirt diameter at base is < at the top
Warn.	20	30	Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	13.5	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	Km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm.
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm.
Distance Upwind of Crest	Lh	0	mm.
Distance from Crest to the Vessel	x	0	mm.
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm.
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Overall M.A.W.P. + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/13
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	SKIRT-1	
Distance "FROM" to "TO"	2000	mm.
Skirt Inside Diameter	6018	mm.
Diameter of Skirt at Base	6016	mm.
Skirt Thickness	22	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-36	
Allowable Stress, Ambient	114	N./mm ²
Allowable Stress, Operating	114	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K02600	
Product Form	Plate; sheet	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	2/13	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	SKIRT-2	
Distance "FROM" to "TO"	1500	mm.
Skirt Inside Diameter	6018	mm.
Diameter of Skirt at Base	6016	mm.
Skirt Thickness	22	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	D	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	3/13	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	BOTTOM HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.



Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	38	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Platform	
Detail ID	PLAT-1	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	360.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	42192	cm ²
Platform Weight	6309.9	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	75.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	3000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	-612.5	mm.
Height/Length of Liquid	662.5	mm.
Liquid Density	0.0009874	kg./cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-2"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	4/13	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	SHELL-1	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars



Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 40
 Detail Type Nozzle
 Detail ID N6-2"
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 900
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 11.279 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 40
 Detail Type Nozzle
 Detail ID K1B-3"
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 900
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 33.688 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element# 5/13
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description SHELL-2
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 6000 mm.
 Element Thickness 32 mm.
 Internal Corrosion Allowance 1.6 mm.
 Nominal Thickness 32 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 13.5 bars
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PLAT-2
 Dist. from "FROM" Node / Offset dist 1000 mm.
 Platform Start Angle (degrees) 270.0
 Platform End Angle (degrees) 90.0
 Platform Wind Area 46256 cm²
 Platform Weight 3247.5 Kgf
 Platform Railing Weight 0 Kg/mm.
 Platform Grating Weight 0.017 Kgs/cm²
 Platform Width 1500 mm.
 Platform Height 1000 mm.
 Platform Clearance or End Offset 85 mm.
 Platform Force Coefficient 1.0



Ladder Layout Angle	90.0	
Ladder Start Elevation	3000	mm.
Ladder End Elevation	7000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	110.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	595.45	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M2-24"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	250.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	595.45	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element#	6/13	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	SHELL-3	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT-3	
Dist. from "FROM" Node / Offset dist	2000	mm.
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	46256	cm ²
Platform Weight	3247.5	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	



Ladder Start Elevation	7000	mm.
Ladder End Elevation	11000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element#	7/13	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	SHELL-4	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PLAT-4	
Dist. from "FROM" Node / Offset dist	3000	mm.
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	46256	cm ²
Platform Weight	3099.5	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	
Ladder Start Elevation	11000	mm.
Ladder End Elevation	15000	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element#	8/13	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	SHELL-5	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element# 9/13
 Element From Node 90
 Element To Node 100
 Element Type Cylinder
 Description SHELL-6
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 6000 mm.
 Element Thickness 32 mm.
 Internal Corrosion Allowance 1.6 mm.
 Nominal Thickness 32 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 13.5 bars
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 90
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 1000 mm.
 Platform Start Angle (degrees) 90.0
 Platform End Angle (degrees) 270.0
 Platform Wind Area 48288 cm²
 Platform Weight 3099.5 Kg
 Platform Railing Weight 0 Kg/mm.
 Platform Grating Weight 0.017 Kgs/cm²
 Platform Width 1500 mm.
 Platform Height 1000 mm.
 Platform Clearance or End Offset 85 mm.
 Platform Force Coefficient 1.0
 Ladder Layout Angle 90.0
 Ladder Start Elevation 15000 mm.
 Ladder End Elevation 19500 mm.
 Unit Weight of Ladder 0 Kg/mm.
 Platform Length (top head platform) 0 mm.

Element# 10/13
 Element From Node 100
 Element To Node 110
 Element Type Cylinder
 Description SHELL-7
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 6000 mm.
 Element Thickness 32 mm.
 Internal Corrosion Allowance 1.6 mm.
 Nominal Thickness 32 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 13.5 bars
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 100
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]



Dist. from "FROM" Node / Offset dist	2000	mm.
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	49507	cm ²
Platform Weight	3099.5	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	
Ladder Start Elevation	19500	mm.
Ladder End Elevation	23500	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element#	11/13	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	SHELL-8	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	K1A-3"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	33.688	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	12/13	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	SHELL-9	
Distance "FROM" to "TO"	1900	mm.
Inside Diameter	6000	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Platform	
Detail ID	PLAT-5	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	46256	cm ²
Platform Weight	3099.5	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	
Ladder Start Elevation	15000	mm.
Ladder End Elevation	19000	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.
Element From Node	120	
Detail Type	Nozzle	
Detail ID	N1-10"	
Dist. from "FROM" Node / Offset dist	950	mm.
Nozzle Diameter	10	in.
Nozzle Schedule	XS	
Nozzle Class	150	
Layout Angle	320.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.485	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	120	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	950	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	50.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	27.667	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	120	
Detail Type	Weight	
Detail ID	N1 PIPING WGT	
Dist. from "FROM" Node / Offset dist	950	mm.
Miscellaneous Weight	2535	Kgf
Offset from Element Centerline	4675	mm.
Element From Node	120	
Detail Type	Weight	
Detail ID	Clip Wgt	
Dist. from "FROM" Node / Offset dist	950	mm.
Miscellaneous Weight	2000	Kgf
Offset from Element Centerline	0	mm.



Element To Node	140
Element Type	Elliptical
Description	TOP HEAD
Distance "FROM" to "TO"	50 mm.
Inside Diameter	6000 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	38 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	13.5 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	130
Detail Type	Platform
Detail ID	SQR PLT
Dist. from "FROM" Node / Offset dist	2138 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	101662 cm ²
Platform Weight	12441 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	8500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	4250 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	15000 mm.
Ladder End Elevation	19100 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	8500 mm.

Element From Node	130
Detail Type	Nozzle
Detail ID	N3-8"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	8 in.
Nozzle Schedule	XS
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.537 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	130
Detail Type	Nozzle
Detail ID	N4-6"
Dist. from "FROM" Node / Offset dist	1250 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	XS
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	54.899 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	130
Detail Type	Nozzle
Detail ID	N5-2"
Dist. from "FROM" Node / Offset dist	1500 mm.



Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	26.271	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Weight	
Detail ID	PSV Weight	
Dist. from "FROM" Node / Offset dist	1800	mm.
Miscellaneous Weight	3000	Kgf
Offset from Element Centerline	4250	mm.
Element From Node	130	
Detail Type	Weight	
Detail ID	N3 Overhead pip	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	1700	Kgf
Offset from Element Centerline	4648	mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
SKIRT-1	...	...	2000	...	2000
SKIRT-2	...	...	3500	...	1500
BOTTOM HEAD	...	...	3550	...	50
SHELL-1	...	...	6550	...	3000
SHELL-2	...	...	9550	...	3000
SHELL-3	...	...	12550	...	3000
SHELL-4	...	...	15550	...	3000
SHELL-5	...	...	18550	...	3000
SHELL-6	...	...	21550	...	3000
SHELL-7	...	...	24550	...	3000
SHELL-8	...	...	27550	...	3000
SHELL-9	...	...	29450	...	1900
TOP HEAD	...	...	29500	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
SKIRT-1	...		22	1	6018	...	1
SKIRT-2	...		22	1	6018	...	2
BOTTOM HEAD	13.564		38	1.6	6000	138	3
SHELL-1	13.5		32	1.6	6000	138	4
SHELL-2	13.5		32	1.6	6000	138	5
SHELL-3	13.5		32	1.6	6000	138	6
SHELL-4	13.5		32	1.6	6000	138	7
SHELL-5	13.5		32	1.6	6000	138	8
SHELL-6	13.5		32	1.6	6000	138	9
SHELL-7	13.5		32	1.6	6000	138	10
SHELL-8	13.5		32	1.6	6000	138	11
SHELL-9	13.5		32	1.6	6000	138	12
TOP HEAD	13.5		38	1.6	6000	138	13

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
SKIRT-1	...	No Calc	No Calc	22	No Calc	1	
SKIRT-2	...	No Calc	No Calc	22	No Calc	2	
BOTTOM HEAD	13.5	13.9073	14.7035	32	31.1129	3	
SHELL-1	13.5	13.8913	14.6255	32	31.1386	4	
SHELL-2	13.5	13.8913	14.6255	32	31.1386	5	
SHELL-3	13.5	13.8913	14.6255	32	31.1386	6	
SHELL-4	13.5	13.8913	14.6255	32	31.1386	7	
SHELL-5	13.5	13.8913	14.6255	32	31.1386	8	
SHELL-6	13.5	13.8913	14.6255	32	31.1386	9	
SHELL-7	13.5	13.8913	14.6255	32	31.1386	10	
SHELL-8	13.5	13.8913	14.6255	32	31.1386	11	
SHELL-9	13.5	13.8913	14.6255	32	31.1386	12	
TOP HEAD	13.5	13.9715	14.7035	32	30.9731	13	

Minimum 14 15

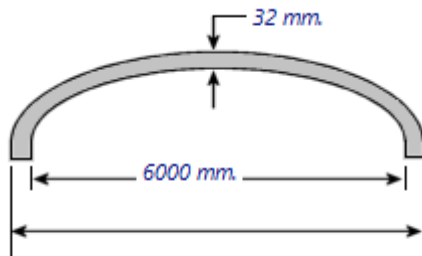
MAWP: 14 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:							
Design Pressure	P	13.564 bars		Int. Design Temperature		82 °C	
Material		SA-516 70		Ext. Design Temperature		0 °C	
Length		50.000 mm.		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.6000 mm.		Allowable Stress (ope)	S	138.0 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm.		Allowable Stress (amb)	Sa	138.0 N./mm ²	
Inside Diameter	Di	6000.000 mm.		Actual Stress		134.0 N./mm ²	
Spec. Min. Thk t _e , or s _a		32.0000 mm.		Spec. Nominal Thk.		38.000 mm.	
Surface Area		409727.4 cm ²		SG of Contents		0.989	
Element Volume		29693370.0 cm ³		Weight of Contents		17566.2 Kg _f	
Empty Weight		13129.8 Kg _f		Operating Weight		30696.0 Kg _f	
				Comp. All. Ope. Stress		110.3 N./mm ²	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (13.564 \cdot 6003.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.564)$$

$$= 29.5129 + 1.6000 = 31.1129 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.064 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 30.4) / (0.999 \cdot 6003.2002 + 0.2 \cdot 30.4)$$

$$= 14 - 0.1 = 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 32.0) / (1.0 \cdot 6000.0 + 0.2 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (13.564 \cdot (0.999 \cdot 6003.2002 + 0.2 \cdot 30.4)) / (2 \cdot 1.0 \cdot 30.4)$$

$$= 133.977 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (13.564 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.564) + 1.6$$

$$= 31.280 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.064 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 36.4) / (3001.6001 + 0.6 \cdot 36.4)$$

$$= 17 - 0.1 = 17 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (6003.2 / (2 \cdot 1501.6))^2) / 6$$

$$= 0.999290$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.743 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 32.0, tr = 29.373, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.966$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 38.0, tr = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.811$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.966	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

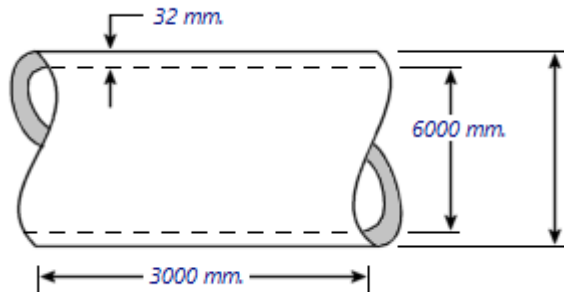
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		14099.1 Kg _f	Operating Weight		14099.1 Kg _f
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



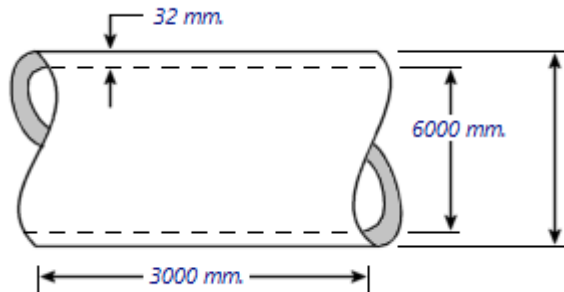
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



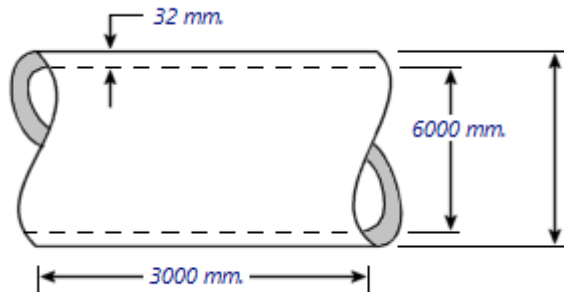
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-3

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



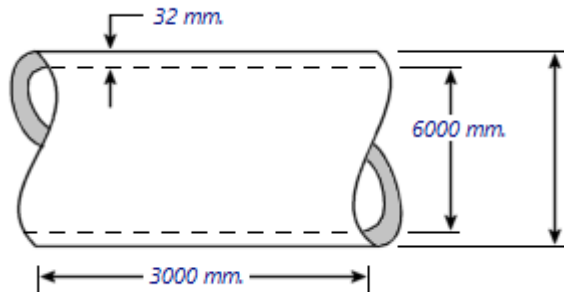
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-4

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm²	SG of Contents		0.000
Element Volume		84838200.0 cm³	Weight of Contents		0.0 Kg _f
Empty Weight		14099.1 Kg _f	Operating Weight		14099.1 Kg _f
			Comp. All. Ope. Stress		110.3 N./mm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



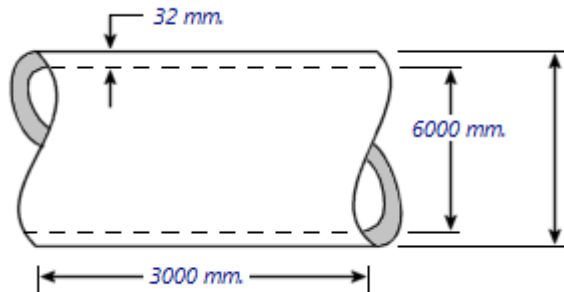
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-5

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



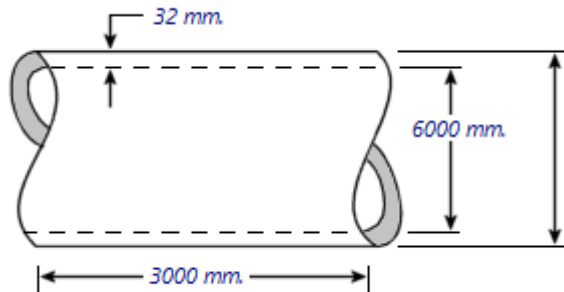
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-6

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



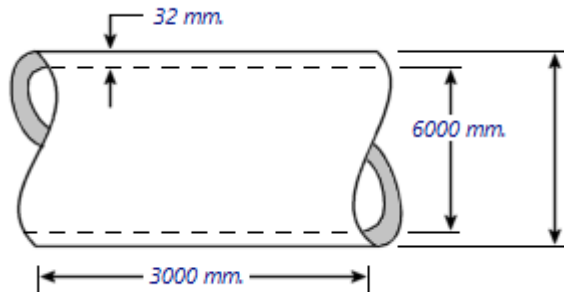
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-7

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



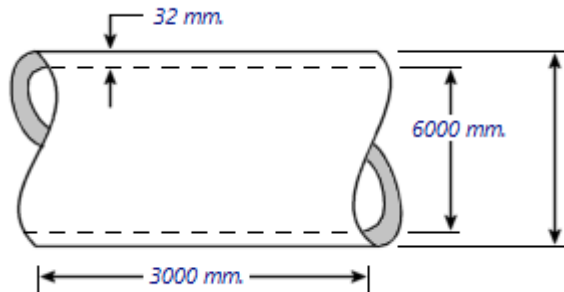
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-8

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		571518.6 cm ²	SG of Contents		0.000
Element Volume		84838200.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		14099.1 Kgf	Operating Weight		14099.1 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



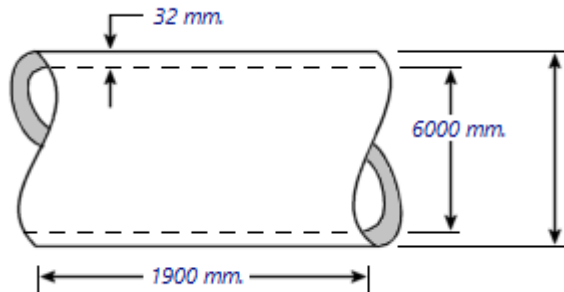
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-9

Cylindrical Shell Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		134.1 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		361961.7 cm ²	SG of Contents		0.000
Element Volume		53730860.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		8929.4 Kgf	Operating Weight		8929.4 Kgf
			Comp. All. Ope. Stress		110.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5)$$

$$= 29.5386 + 1.6000 = 31.1386 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (3001.6001 + 0.6 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 32.0) / (3000.0 + 0.6 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13.5 \cdot (3001.6001 + 0.6 \cdot 30.4)) / (1.0 \cdot 30.4)$$

$$= 134.113 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.531 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32.0, t_r = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.972, Temp. Reduction = 2 °C



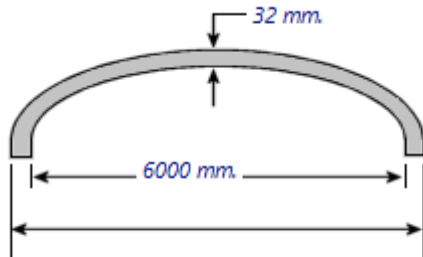
Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.972	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:					
Design Pressure	P	13.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	6000.000 mm.	Actual Stress		133.3 N./mm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm.	Spec. Nominal Thk.		38.000 mm.
Surface Area		409727.4 cm ²	SG of Contents		0.000
Element Volume		29693370.0 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		13129.8 Kg _f	Operating Weight		13129.8 Kg _f
			Comp. All. Ope. Stress		110.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (13.5 \cdot 6003.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$$

$$= 29.3731 + 1.6000 = 30.9731 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 30.4) / (0.999 \cdot 6003.2002 + 0.2 \cdot 30.4)$$

$$= 14 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 32.0) / (1.0 \cdot 6000.0 + 0.2 \cdot 32.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (13.5 \cdot (0.999 \cdot 6003.2002 + 0.2 \cdot 30.4)) / (2 \cdot 1.0 \cdot 30.4)$$

$$= 133.343 \text{ N./mm}^2$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (13.5 \cdot 3001.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 13.5) + 1.6$$

$$= 31.139 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 36.4) / (3001.6001 + 0.6 \cdot 36.4)$$

$$= 17 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (6003.2 / (2 \cdot 1501.6))^2) / 6$$

$$= 0.999290$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.743 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 32.0$, $t_r = 29.373$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.966$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 38.0$, $t_r = 29.539$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.811$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	13.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.966	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.550 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	17.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	18.425 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.850 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	19.305 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	19.305 bars
App 27-4	= M.A.W.P.	13.500 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	178.4	234.0	0.763	19.01
SHELL-1	179.4	234.0	0.767	19.01
SHELL-2	179.4	234.0	0.767	19.01
SHELL-3	179.4	234.0	0.767	19.01
SHELL-4	179.4	234.0	0.767	19.01
SHELL-5	179.4	234.0	0.767	19.01
SHELL-6	179.4	234.0	0.767	19.01
SHELL-7	179.4	234.0	0.767	19.01
SHELL-8	179.4	234.0	0.767	19.01
SHELL-9	179.4	234.0	0.767	19.01
TOP HEAD	178.4	234.0	0.763	19.01

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K1B-3"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000
K1A-3"	Nozzle	138.00	138.00	1.000
N1-10"	Nozzle	118.00	118.00	1.000
N1-10"	Pad	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N3-8"	Nozzle	118.00	118.00	1.000
N3-8"	Pad	138.00	138.00	1.000
N4-6"	Nozzle	118.00	118.00	1.000
N4-6"	Pad	138.00	138.00	1.000
N5-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
SHELL-3	138.00	138.00	1.000
SHELL-4	138.00	138.00	1.000
SHELL-5	138.00	138.00	1.000
SHELL-6	138.00	138.00	1.000
SHELL-7	138.00	138.00	1.000
SHELL-8	138.00	138.00	1.000



SHELL-9	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-2"	4.02	225.00	0.018
N6-2"	2.93	225.00	0.013
K1B-3"	3.99	225.00	0.018
M1-24"	30.12	234.00	0.129
M2-24"	30.12	234.00	0.129
K1A-3"	3.99	225.00	0.018
N1-10"	19.68	216.00	0.091
K2-2"	2.93	225.00	0.013
N3-8"	15.64	216.00	0.072
N4-6"	13.82	216.00	0.064
N5-2"	4.02	225.00	0.018

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.550 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	17.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	18.675 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.850 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	19.305 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	19.305 bars
App 27-4	= M.A.W.P.	13.500 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 13.5 • 1.0
 = 17.550 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	201.4	234.0	0.861	20.39
SHELL-1	201.1	234.0	0.859	20.24
SHELL-2	198.2	234.0	0.847	19.95
SHELL-3	195.2	234.0	0.834	19.65
SHELL-4	192.3	234.0	0.822	19.36
SHELL-5	189.4	234.0	0.809	19.06
SHELL-6	186.5	234.0	0.797	18.77
SHELL-7	183.5	234.0	0.784	18.48
SHELL-8	180.6	234.0	0.772	18.18
SHELL-9	177.7	234.0	0.759	17.89
TOP HEAD	174.8	234.0	0.747	17.70

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K1B-3"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000
K1A-3"	Nozzle	138.00	138.00	1.000
N1-10"	Nozzle	118.00	118.00	1.000
N1-10"	Pad	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N3-8"	Nozzle	118.00	118.00	1.000
N3-8"	Pad	138.00	138.00	1.000
N4-6"	Nozzle	118.00	118.00	1.000
N4-6"	Pad	138.00	138.00	1.000
N5-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
SHELL-3	138.00	138.00	1.000



SHELL-4	138.00	138.00	1.000
SHELL-5	138.00	138.00	1.000
SHELL-6	138.00	138.00	1.000
SHELL-7	138.00	138.00	1.000
SHELL-8	138.00	138.00	1.000
SHELL-9	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-2"	4.31	225.00	0.019
N6-2"	3.12	225.00	0.014
K1B-3"	4.25	225.00	0.019
M1-24"	31.60	234.00	0.135
M2-24"	31.60	234.00	0.135
K1A-3"	3.82	225.00	0.017
N1-10"	18.52	216.00	0.086
K2-2"	2.76	225.00	0.012
N3-8"	14.56	216.00	0.067
N4-6"	12.87	216.00	0.060
N5-2"	3.74	225.00	0.017

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3500	...	...	No Calc	No Calc
20	30	3500	...	...	No Calc	No Calc
30	40	No Calc	6064	30.4	0.00069628	69.612
40	50	27000	6064	30.4	0.98477E-04	9.84547
50	60	27000	6064	30.4	0.98477E-04	9.84547
60	70	27000	6064	30.4	0.98477E-04	9.84547
70	80	27000	6064	30.4	0.98477E-04	9.84547
80	90	27000	6064	30.4	0.98477E-04	9.84547
90	100	27000	6064	30.4	0.98477E-04	9.84547
100	110	27000	6064	30.4	0.98477E-04	9.84547
110	120	27000	6064	30.4	0.98477E-04	9.84547
120	130	27000	6064	30.4	0.98477E-04	9.84547
130	140	No Calc	6064	30.4	0.00069628	69.612

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	32	No Calc	...	3.87731
40	50	32	No Calc	...	0.65806
50	60	32	No Calc	...	0.65806
60	70	32	No Calc	...	0.65806
70	80	32	No Calc	...	0.65806
80	90	32	No Calc	...	0.65806
90	100	32	No Calc	...	0.65806
100	110	32	No Calc	...	0.65806
110	120	32	No Calc	...	0.65806
120	130	32	No Calc	...	0.65806
130	140	32	No Calc	...	3.87731

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3500	No Calc	No Calc	No Calc
20	30	3500	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	27000	No Calc	No Calc	No Calc
50	60	27000	No Calc	No Calc	No Calc
60	70	27000	No Calc	No Calc	No Calc
70	80	27000	No Calc	No Calc	No Calc
80	90	27000	No Calc	No Calc	No Calc
90	100	27000	No Calc	No Calc	No Calc
100	110	27000	No Calc	No Calc	No Calc
110	120	27000	No Calc	No Calc	No Calc
120	130	27000	No Calc	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.





Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	10060.7	...	9914.17	...	704.247
20	30	4852.15	...	4742.3	...	339.651
30	40	13129.8	29693370	12576.9	29740154	919.083
40	50	14099.1	84838200	13397.6	84928720	986.935
50	60	14099.1	84838200	13397.6	84928720	986.935
60	70	14099.1	84838200	13397.6	84928720	986.935
70	80	14099.1	84838200	13397.6	84928720	986.935
80	90	14099.1	84838200	13397.6	84928720	986.935
90	100	14099.1	84838200	13397.6	84928720	986.935
100	110	14099.1	84838200	13397.6	84928720	986.935
110	120	14099.1	84838200	13397.6	84928720	986.935
120	130	8929.41	53730860	8485.16	53788192	625.059
130	140	13129.8	29693370	12576.9	29740154	919.083
Total		162894	0.792E+09	155476	0.793E+09	11402

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	6751.64	...	...	0.34391E-04	PLAT-1
30	Liqd	17566.2	...	-331.25	...	Liquid: 20
30	Nozl	10.7002	...	-1614	...	N2-2"
40	Nozl	12.0686	-3093.66	1500	...	N6-2"
40	Nozl	36.0459	3130	1500	...	K1B-3"
50	Plat	3474.79	...	1000	2472.56	PLAT-2
50	Nozl	637.129	2983.53	1500	1085.92	M1-24"
50	Nozl	637.129	-2983.53	1500	1085.92	M2-24"
60	Plat	3474.79	...	2000	-2472.56	PLAT-3
70	Plat	3316.43	...	3000	2472.56	PLAT-4
90	Plat	3316.43	...	1000	-2472.56	PLAT:[1 OF 1]
100	Plat	3316.43	...	2000	2472.56	PLAT:[1 OF 1]
110	Nozl	36.0459	3130	1500	...	K1A-3"
120	Plat	3316.43	...	...	-2472.56	PLAT-5
120	Nozl	79.6992	-2011.92	950	-2397.72	N1-10"
120	Nozl	29.6034	2397.72	950	-2011.92	K2-2"
120	Wght	2535	-3005.03	950.002	-3581.26	N1 PIPING WGT
120	Wght	2000	...	950.002	...	Clip Wgt
130	Plat	13311.4	...	2138	...	SQR PLT
130	Nozl	106.505	...	1759.01	...	N3-8"
130	Nozl	58.742	-1250	1672.59	...	N4-6"
130	Nozl	28.1105	1500	1708.04	...	N5-2"
130	Wght	3000	3005.2	1800	-3005.2	PSV Weight
130	Wght	1700	4648	...	...	N3 Overhead pip
120	Trun	1172.87	...	950	...	Trunnions

Total Weight of Each Detail Type:

Platforms	40278.3
Liquid	17566.2
Nozzles	1671.8
Weights	9235.0
Trunnions	1172.9

Sum of the Detail Weights 69924.2 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
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Main Elements	174296.9	174296.9	174296.9	174296.9	174296.9	174296.9
Nozzles	1671.8	1671.8	1671.8	1671.8	1671.8	1671.8
Wld Weights	2000.0	2000.0	2000.0	2000.0	2000.0	2000.0
Trunnions	1172.9	1172.9	1172.9	1172.9	1172.9	1172.9
Platforms	...	...	...	40278.3	40278.3	40278.3
Empty Weights	...	...	...	7235.0	7235.0	...
Test Weights	...	7235.0	...	...	...	...
Ope Weights	...	...	...	...	...	7235.0
Ope. Liquid	...	...	...	...	...	17566.2
Test Liquid	...	791339.9	...	...	...	...
Totals	179141.5	977716.5	179141.5	226654.9	226654.9	244221.1

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	179141.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	977716.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	179141.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	226654.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	226654.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	244221.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	1017994.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	89724.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	389513.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	92763.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	380887	378122
20	30	285665	283592
30	40	409727	399660
40	50	571519	565487
50	60	571519	565487
60	70	571519	565487
70	80	571519	565487
80	90	571519	565487
90	100	571519	565487
100	110	571519	565487
110	120	571519	565487
120	130	361962	358142
130	140	409727	399660

Total	6420117.0	6343069.0 cm ²
Total	6910.6	6827.6 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	10764.9	10764.9	10764.9	...	10764.9
20	30	5191.8	5191.8	5191.8	...	5191.8
30	40	14059.5	38377.4	50486.4	0.0002322	20811.2



40	50	15134.1	15134.1	99920.5	...	15134.1
50	60	16360.3	19835.1	104621	9975.58	19835.1
60	70	15086	18560.8	103347	8591.81	18560.8
70	80	15086	18402.4	103189	8200.25	18402.4
80	90	15086	15086	99872.4	...	15086
90	100	15086	18402.4	103189	8200.25	18402.4
100	110	15086	18402.4	103189	8200.25	18402.4
110	120	15122	15122	99908.5	...	15122
120	130	15371.6	18688.1	72386.1	17529.6	18688.1
130	140	18942.2	32253.6	61928.8	9015.79	32253.6

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	226655	244221	1017994
20	30	215890	233456	1007229
30	40	210698	228264	1002037
40	50	189887	189887	951552
50	60	174753	174753	851631
60	70	154918	154918	747010
70	80	136357	136357	643662
80	90	117955	117955	540474
90	100	102869	102869	440601
100	110	84466.1	84466.1	337412
110	120	66063.7	66063.7	234223
120	130	50941.6	50941.6	134315
130	140	32253.6	32253.6	61928.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	69713.5	69713.5	69713.5
20	30	69713.5	69713.5	69713.5
30	40	69713.5	69713.5	69713.5
40	50	69713.5	69713.5	69713.5
50	60	69713.5	69713.5	69713.5
60	70	59737.9	59737.9	59737.9
70	80	51146.1	51146.1	51146.1
80	90	42945.9	42945.9	42945.9
90	100	42945.9	42945.9	42945.9
100	110	34745.6	34745.6	34745.6
110	120	26545.4	26545.4	26545.4
120	130	26545.4	26545.4	26545.4
130	140	9015.79	9015.79	9015.79



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N6-2"	143.41	153.10	82	900	GR 1.1	...	...	...	...
K1B-3"	143.41	153.10	82	900	GR 1.1	...	...	...	...
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
M2-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K1A-3"	143.41	153.10	82	900	GR 1.1	...	...	...	...
N1-10"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K2-2"	143.41	153.10	82	900	GR 1.1	...	...	...	...
N3-8"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N4-6"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N5-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 3.90801 Hz.

The Natural Frequency for the Vessel (Ope...) is 3.90729 Hz.

The Natural Frequency for the Vessel (Filled) is 1.96586 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.182 \cdot 0.9) / (1 + 1.7 \cdot 3.4 \cdot 0.182)))$
 $= \min(0.85, 0.878)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	3.907 Hz
Natural Frequency of Vessel (Empty)	3.908 Hz
Natural Frequency of Vessel (Test)	1.966 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.570
Structure Height to Diameter ratio	5.181
Height to top of Structure	31032.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.055 + 300.0 = 1300.055$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.265 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.57 \cdot 156.576$$

$$= 1213.1 \text{ lbs. } [550.3] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
SKIRT-1	1300.1	0.000	0.000	0.000	0.849	1.000	78.120
SKIRT-2	3050.0	0.000	0.000	0.000	0.849	1.000	78.120
BOTTOM HEAD	3825.0	0.000	0.000	0.000	0.849	1.000	78.120
SHELL-1	5350.0	0.000	0.000	0.000	0.877	1.000	78.120
SHELL-2	8350.0	0.000	0.000	0.000	0.964	1.000	78.120
SHELL-3	11350.0	0.000	0.000	0.000	1.028	1.000	78.120
SHELL-4	14350.0	0.000	0.000	0.000	1.080	1.000	78.120
SHELL-5	17350.0	0.000	0.000	0.000	1.124	1.000	78.120
SHELL-6	20350.0	0.000	0.000	0.000	1.162	1.000	78.120
SHELL-7	23350.0	0.000	0.000	0.000	1.197	1.000	78.120
SHELL-8	26350.0	0.000	0.000	0.000	1.227	1.000	78.120
SHELL-9	28800.0	0.000	0.000	0.000	1.251	1.000	78.120
TOP HEAD	30423.3	0.000	0.000	0.000	1.265	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(10⁴)
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0.0/29500.0 = 0.0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)

:: [19.89/96.78²] • 10⁴ = 21.238 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.400E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 236284 / (29500.0 \cdot 6064.0^2)$$



= 0.21782E-06

Compute the damping factor Df which is a measure of instability [Df]:

$$= W \cdot DI / (L \cdot Dr^2)$$

$$= 236284 \cdot 0.03 / (29500.0 \cdot 6064.0^2)$$

$$= 0.65345E-08$$

Compute the critical wind velocity [Vc]:

$$= 3.4 \cdot f \cdot Dr$$

$$= 3.4 \cdot 3.907 \cdot 19.895$$

$$= 264.301 \text{ mile/hr } [425.340 \text{ Km/hr}]$$

Compute the velocity at the top of the tower [Vw]:

$$= V30 \cdot (L / (30 + \text{BaseHeight}))^{0.143}$$

$$= 60.4 \cdot (96.78 / (30 + 1.0))^{0.143}$$

$$= 71.083 \text{ mile/hr } [114.394 \text{ Km/hr}]$$

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

$$= Vw \cdot Gh$$

$$= 114.394 \cdot 0.85$$

$$= 97.235 \text{ Km/hr}$$

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT-1	42192.00	3800.00	78.12	329.60	1.00
PLAT-2	46256.00	7850.00	78.12	361.35	1.00
PLAT-3	46256.00	11850.00	78.12	361.35	1.00
PLAT-4	46256.00	15850.00	78.12	361.35	1.00
PLAT:[1 OF 1]	48288.00	19850.00	78.12	377.23	1.00
PLAT:[1 OF 1]	49507.20	23850.00	78.12	386.75	1.00
PLAT-5	46256.00	27850.00	78.12	361.35	1.00
SQR PLT	101662.40	31888.00	78.12	794.19	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
N1 PIPING WGT	0.00	28800.00	78.12	0.00
Clip Wgt	0.00	28800.00	78.12	0.00
PSV Weight	0.00	31550.00	78.12	0.00
N3 Overhead p	0.00	29750.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 3.90729 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300.06	7273.2	145464	78.12	605.289
20	30	3050.04	7273.2	109098	78.12	453.967
30	40	3825	7276.8	3638.4	78.12	375.335
40	50	5350	7276.8	218304	78.12	908.383
50	60	8350	7276.8	218304	78.12	1282.07
60	70	11350	7276.8	218304	78.12	1323.38
70	80	14350	7276.8	218304	78.12	1347.42
80	90	17350	7276.8	218304	78.12	908.383
90	100	20350	7276.8	218304	78.12	1313.14
100	110	23350	7276.8	218304	78.12	1342.92
110	120	26350	7276.8	218304	78.12	908.383



120	130	28800	7276.8	138259	78.12	959.686
130	140	30423.3	7276.8	91195	78.12	1295.14

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(101.706^{0.75})$
 = 0.641 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/3.907)
 = 0.256

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.641$, 0.256) per 12.8.2
 = 0.256

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.157 / (1.25 / 1.25)$
 = 0.157

Maximum value of Cs per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.104 / (0.256(1.25 / 1.25))$
 = 0.406

Minimum value of Cs per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.1568 \cdot 244221.1$
 = 38293.863 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 244221$

= 7658.8 Kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x / k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 3.90729 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	10764.9	108.836	106.312	7658.77
20	30	2750	5191.8	144.348	141.001	7321.18
30	40	3525	38377.4	1367.71	489.441	7158.37
40	50	5050	15134.1	772.695	754.776	5954.85
50	60	8050	19835.1	1614.32	1300.64	5480.25
60	70	11050	18560.8	2073.57	1646.29	4858.22
70	80	14050	18402.4	2614.03	2093.25	4276.15
80	90	17050	15086	2600.51	2540.2	3699.05
90	100	20050	18402.4	3730.35	2987.16	3225.96
100	110	23050	18402.4	4288.5	3434.12	2648.86
110	120	26050	15122	3982.7	3890.34	2071.76
120	130	28500	18688.1	5384.79	4326.48	1597.53
130	140	29475	32253.6	9611.5	5513.84	1011.47



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg-f	Earthquake Shear Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	13023.5	38293.9	214442	846609
20	30	2750	12418.2	38185	189000	770129
30	40	3525	11964.2	38040.7	170713	712958
40	50	5050	11588.9	36673	170124	711090
50	60	8050	10680.5	35900.3	136719	602228
60	70	11050	9398.45	34286	106600	496947
70	80	14050	8075.07	32212.4	80389.6	397198
80	90	17050	6727.65	29598.4	58185.1	304480
90	100	20050	5819.26	26997.8	39364.4	219584
100	110	23050	4506.13	23267.5	23876	144184
110	120	26050	3163.21	18979	12371.7	80813
120	130	28500	2254.82	14996.3	4244.51	29849
130	140	29475	1295.14	9611.5	871.984	6471.18

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	13023.5	0.0027805	0.010896	0.10672E-04
20	30	12418.2	0.020284	0.032348	0.17809E-04
30	40	11964.2	0.032794	0.033242	0.17969E-04
40	50	11588.9	0.063667	0.10056	0.26596E-04
50	60	10680.5	0.14323	0.19102	0.33430E-04
60	70	9398.45	0.24332	0.29955	0.38676E-04
70	80	8075.07	0.35918	0.42172	0.42559E-04
80	90	6727.65	0.48671	0.55376	0.45293E-04
90	100	5819.26	0.62247	0.6925	0.47055E-04
100	110	4506.13	0.76354	0.83533	0.48057E-04
110	120	3163.21	0.90765	0.9803	0.48512E-04
120	130	2254.82	1.02642	1.07257	0.48598E-04
130	140	1295.14	1.07379	1.075	0.48598E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	510.155	3188.47
20	30	510.155	3188.47
30	40	510.407	3190.05
40	50	510.407	3190.05
50	60	510.407	3190.05
60	70	510.407	3190.05
70	80	510.407	3190.05
80	90	510.407	3190.05
90	100	510.407	3190.05
100	110	510.407	3190.05
110	120	510.407	3190.05
120	130	510.407	3190.05
130	140	510.407	3190.05

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 147.500 Actual deflection : 1.075 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	4173.17	3984.15	627662144	599429376
20	30	4173.17	3984.15	627662144	599429376
30	40	6064.02	5762.33	909652928	864853696
40	50	6064.02	5762.33	909652928	864853696
50	60	6064.02	5762.33	909652928	864853696
60	70	6064.02	5762.33	909652928	864853696
70	80	6064.02	5762.33	909652928	864853696
80	90	6064.02	5762.33	909652928	864853696
90	100	6064.02	5762.33	909652928	864853696
100	110	6064.02	5762.33	909652928	864853696
110	120	6064.02	5762.33	909652928	864853696
120	130	6064.02	5762.33	909652928	864853696
130	140	6064.02	5762.33	909652928	864853696

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-96.9195	-96.9195
20	30	82.8	140.4	-96.9195	-96.9195
30	40	165.6	280.8	-110.326	-110.326
40	50	165.6	280.8	-110.326	-110.326
50	60	165.6	280.8	-110.326	-110.326
60	70	165.6	280.8	-110.326	-110.326
70	80	165.6	280.8	-110.326	-110.326
80	90	165.6	280.8	-110.326	-110.326
90	100	165.6	280.8	-110.326	-110.326
100	110	165.6	280.8	-110.326	-110.326
110	120	165.6	280.8	-110.326	-110.326
120	130	165.6	280.8	-110.326	-110.326
130	140	165.6	280.8	-110.326	-110.326

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	66.3812	...	86.2956
40	50	66.3812	...	86.2956
50	60	66.3812	...	86.2956
60	70	66.3812	...	86.2956
70	80	66.3812	...	86.2956
80	90	66.3812	...	86.2956
90	100	66.3812	...	86.2956
100	110	66.3812	...	86.2956
110	120	66.3812	...	86.2956
120	130	66.3812	...	86.2956
130	140	66.3812	...	86.2956

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-5.57903	-6.01141	-25.0576	1.14051	1.14051
20	30	-5.31405	-5.74644	-24.7926	1.14051	1.14051
30	40	-3.58584	-3.58584	-3.58584	0.79049	0.79049
40	50	-3.23166	-3.23166	-3.23166	0.79049	0.79049
50	60	-2.97409	-2.97409	-2.97409	0.79049	0.79049
60	70	-2.63652	-2.63652	-2.63652	0.67737	0.67737
70	80	-2.32064	-2.32064	-2.32064	0.57995	0.57995
80	90	-2.00745	-2.00745	-2.00745	0.48697	0.48697
90	100	-1.7507	-1.7507	-1.7507	0.48697	0.48697
100	110	-1.43752	-1.43752	-1.43752	0.39398	0.39398
110	120	-1.12433	-1.12433	-1.12433	0.301	0.301
120	130	-0.86697	-0.86697	-0.86697	0.301	0.301
130	140	-0.54892	-0.54892	-0.54892	0.10223	0.10223

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Opr. Wind N./mm ²	Bend. Str. Opr. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	1.14051	3.50827	13.8505	1.75413	...
20	30	1.14051	3.09204	12.5993	1.54602	...
30	40	0.79049	1.93573	8.08428	0.96786	...
40	50	0.79049	1.92905	8.0631	0.96453	...
50	60	0.79049	1.55027	6.82871	0.77514	...
60	70	0.67737	1.20875	5.63492	0.60437	...
70	80	0.57995	0.91154	4.50385	0.45577	...
80	90	0.48697	0.65976	3.45251	0.32988	...
90	100	0.48697	0.44636	2.48987	0.22318	...
100	110	0.39398	0.27073	1.63491	0.13537	...
110	120	0.301	0.14028	0.91634	0.070142	...
120	130	0.301	0.048129	0.33846	0.024064	...
130	140	0.10223	0.0098875	0.073377	0.0049437	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	10.4249
20	30	...	...	...	9.47037
30	40	...	...	...	6.06985
40	50	...	...	...	6.05356
50	60	...	...	...	5.09732
60	70	...	...	...	4.17605
70	80	...	...	...	3.3049
80	90	...	...	...	2.49736
90	100	...	...	...	1.76863
100	110	...	...	...	1.13391
110	120	...	...	...	0.60841
120	130	...	...	...	0.20749
130	140	...	...	...	0.042094

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.18852
20	30	...	0.18021
30	40	...	0.12183
40	50	...	0.10134
50	60	...	0.093268
60	70	...	0.082681
70	80	...	0.072775
80	90	...	0.062954
90	100	...	0.054902
100	110	...	0.045081
110	120	...	0.035259
120	130	...	0.027188
130	140	...	0.017214

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...
130	140	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
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10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-10.23	96.92		0.1055
20		82.80	-9.55	96.92		0.0985
30		165.60	-6.31	110.33		0.0572
40		165.60	-5.95	110.33		0.0539
50		165.60	-5.31	110.33		0.0482
60		165.60	-4.52	110.33		0.0410
70		165.60	-3.81	110.33		0.0346
80		165.60	-3.15	110.33		0.0286
90		165.60	-2.68	110.33		0.0243
100		165.60	-2.10	110.33		0.0191
110		165.60	-1.57	110.33		0.0142
120		165.60	-1.22	110.33		0.0110
130		165.60	-0.66	110.33		0.0060

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	95.76	-16.96	96.92	0.0645	0.1749
20	5.48	82.80	-15.74	96.92	0.0661	0.1625
30	3.40	165.60	-10.32	110.33	0.0205	0.0936
40	3.71	165.60	-9.97	110.33	0.0224	0.0904
50	3.01	165.60	-8.77	110.33	0.0182	0.0795
60	2.30	165.60	-7.41	110.33	0.0139	0.0671



70	1.64	165.60	-6.13	110.33	0.0099	0.0556
80	1.04	165.60	-4.93	110.33	0.0063	0.0447
90	0.56	165.60	-3.95	110.33	0.0034	0.0358
100	0.14	165.60	-2.92	110.33	0.0008	0.0265
110		165.60	-2.00	110.33		0.0181
120		165.60	-1.35	110.33		0.0122
130		165.60	-0.68	110.33		0.0061

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-10.66	96.92		0.1100
20		82.80	-9.98	96.92		0.1030
30		165.60	-6.31	110.33		0.0572
40		165.60	-5.95	110.33		0.0539
50		165.60	-5.31	110.33		0.0482
60		165.60	-4.52	110.33		0.0410
70		165.60	-3.81	110.33		0.0346
80		165.60	-3.15	110.33		0.0286
90		165.60	-2.68	110.33		0.0243
100		165.60	-2.10	110.33		0.0191
110		165.60	-1.57	110.33		0.0142
120		165.60	-1.22	110.33		0.0110
130		165.60	-0.66	110.33		0.0060

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.17	95.76	-20.81	96.92	0.0957	0.2148
20	8.17	82.80	-19.31	96.92	0.0987	0.1992
30	5.41	165.60	-12.34	110.33	0.0327	0.1118
40	5.72	165.60	-11.98	110.33	0.0346	0.1086
50	4.74	165.60	-10.50	110.33	0.0286	0.0952
60	3.76	165.60	-8.87	110.33	0.0227	0.0804
70	2.84	165.60	-7.33	110.33	0.0171	0.0665
80	1.99	165.60	-5.88	110.33	0.0120	0.0533
90	1.28	165.60	-4.67	110.33	0.0077	0.0424
100	0.64	165.60	-3.42	110.33	0.0038	0.0310
110	0.13	165.60	-2.31	110.33	0.0008	0.0209
120		165.60	-1.48	110.33		0.0134
130		165.60	-0.71	110.33		0.0064

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-27.95	96.92		0.2884
20		140.40	-27.48	96.92		0.2835
30		280.80	-5.34	110.33		0.0484
40		280.80	-4.99	110.33		0.0452
50		280.80	-4.54	110.33		0.0411
60		280.80	-3.92	110.33		0.0355
70		280.80	-3.36	110.33		0.0304
80		280.80	-2.82	110.33		0.0256
90		280.80	-2.46	110.33		0.0223
100		280.80	-1.97	110.33		0.0178
110		280.80	-1.50	110.33		0.0136
120		280.80	-1.19	110.33		0.0108
130		280.80	-0.66	110.33		0.0059

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-26.20	96.92		0.2703
20		140.40	-25.93	96.92		0.2676
30		280.80	-4.38	110.33		0.0397
40		280.80	-4.02	110.33		0.0365



50	280.80	-3.76	110.33	0.0341
60	280.80	-3.31	110.33	0.0300
70	280.80	-2.90	110.33	0.0263
80	280.80	-2.49	110.33	0.0226
90	280.80	-2.24	110.33	0.0203
100	280.80	-1.83	110.33	0.0166
110	280.80	-1.43	110.33	0.0129
120	280.80	-1.17	110.33	0.0106
130	280.80	-0.65	110.33	0.0059

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-10.66	96.91		0.1100
20		82.80	-9.98	96.91		0.1030
30	65.52	165.60		110.33	0.3957	
40	65.87	165.60		110.33	0.3978	
50	65.75	165.60		110.33	0.3970	
60	65.63	165.60		110.33	0.3963	
70	65.55	165.60		110.33	0.3958	
80	65.52	165.60		110.33	0.3957	
90	65.56	165.60		110.33	0.3959	
100	65.61	165.60		110.33	0.3962	
110	65.70	165.60		110.33	0.3967	
120	65.86	165.60		110.33	0.3977	
130	65.94	165.60		110.33	0.3982	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.17	95.76	-20.81	96.91	0.0957	0.2148
20	8.17	82.80	-19.31	96.91	0.0987	0.1992
30	71.79	165.60		110.33	0.4335	
40	72.10	165.60		110.33	0.4354	
50	71.12	165.60		110.33	0.4295	
60	70.14	165.60		110.33	0.4235	
70	69.22	165.60		110.33	0.4180	
80	68.38	165.60		110.33	0.4129	
90	67.66	165.60		110.33	0.4086	
100	67.02	165.60		110.33	0.4047	
110	66.51	165.60		110.33	0.4016	
120	66.18	165.60		110.33	0.3996	
130	66.03	165.60		110.33	0.3987	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-10.66	96.91		0.1100
20		82.80	-9.98	96.91		0.1030
30		165.60	-6.31	110.33		0.0572
40		165.60	-5.95	110.33		0.0539
50		165.60	-5.31	110.33		0.0482
60		165.60	-4.52	110.33		0.0410
70		165.60	-3.81	110.33		0.0346
80		165.60	-3.15	110.33		0.0286
90		165.60	-2.68	110.33		0.0243
100		165.60	-2.10	110.33		0.0191
110		165.60	-1.57	110.33		0.0142
120		165.60	-1.22	110.33		0.0110
130		165.60	-0.66	110.33		0.0060

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.17	95.76	-20.81	96.91	0.0957	0.2148
20	8.17	82.80	-19.31	96.91	0.0987	0.1992



30	5.41	165.60	-12.34	110.33	0.0327	0.1118
40	5.72	165.60	-11.98	110.33	0.0346	0.1086
50	4.74	165.60	-10.50	110.33	0.0286	0.0952
60	3.76	165.60	-8.87	110.33	0.0227	0.0804
70	2.84	165.60	-7.33	110.33	0.0171	0.0665
80	1.99	165.60	-5.88	110.33	0.0120	0.0533
90	1.28	165.60	-4.67	110.33	0.0077	0.0424
100	0.64	165.60	-3.42	110.33	0.0038	0.0310
110	0.13	165.60	-2.31	110.33	0.0008	0.0209
120		165.60	-1.48	110.33		0.0134
130		165.60	-0.71	110.33		0.0064

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-27.95	96.92		0.2884
20		140.40	-27.48	96.92		0.2835
30	84.47	280.80		110.33	0.3008	
40	84.82	280.80		110.33	0.3021	
50	84.89	280.80		110.33	0.3023	
60	84.94	280.80		110.33	0.3025	
70	85.01	280.80		110.33	0.3027	
80	85.10	280.80		110.33	0.3031	
90	85.26	280.80		110.33	0.3036	
100	85.39	280.80		110.33	0.3041	
110	85.54	280.80		110.33	0.3046	
120	85.75	280.80		110.33	0.3054	
130	85.85	280.80		110.33	0.3057	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-26.20	96.92		0.2703
20		140.40	-25.93	96.92		0.2676
30	83.50	280.80		110.33	0.2974	
40	83.85	280.80		110.33	0.2986	
50	84.11	280.80		110.33	0.2995	
60	84.34	280.80		110.33	0.3003	
70	84.55	280.80		110.33	0.3011	
80	84.78	280.80		110.33	0.3019	
90	85.03	280.80		110.33	0.3028	
100	85.25	280.80		110.33	0.3036	
110	85.47	280.80		110.33	0.3044	
120	85.73	280.80		110.33	0.3053	
130	85.85	280.80		110.33	0.3057	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-6.72	96.91		0.0693
20		82.80	-6.45	96.91		0.0666
30	63.59	165.60		110.33	0.3840	
40	63.94	165.60		110.33	0.3861	
50	64.20	165.60		110.33	0.3877	
60	64.42	165.60		110.33	0.3890	
70	64.64	165.60		110.33	0.3903	
80	64.86	165.60		110.33	0.3917	
90	65.12	165.60		110.33	0.3932	
100	65.34	165.60		110.33	0.3946	
110	65.56	165.60		110.33	0.3959	
120	65.82	165.60		110.33	0.3974	
130	65.93	165.60		110.33	0.3982	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		95.76	-5.74	96.91	0.0592
20		82.80	-5.49	96.91	0.0566
30	62.97	165.60		110.33	0.3803
40	63.31	165.60		110.33	0.3823
50	63.56	165.60		110.33	0.3838
60	63.88	165.60		110.33	0.3857
70	64.18	165.60		110.33	0.3875
80	64.47	165.60		110.33	0.3893
90	64.72	165.60		110.33	0.3908
100	65.02	165.60		110.33	0.3926
110	65.31	165.60		110.33	0.3944
120	65.56	165.60		110.33	0.3959
130	65.86	165.60		110.33	0.3977

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.15	96.91		0.0738
20		82.80	-6.89	96.91		0.0711
30	63.59	165.60		110.33	0.3840	
40	63.94	165.60		110.33	0.3861	
50	64.20	165.60		110.33	0.3877	
60	64.42	165.60		110.33	0.3890	
70	64.64	165.60		110.33	0.3903	
80	64.86	165.60		110.33	0.3917	
90	65.12	165.60		110.33	0.3932	
100	65.34	165.60		110.33	0.3946	
110	65.56	165.60		110.33	0.3959	
120	65.82	165.60		110.33	0.3974	
130	65.93	165.60		110.33	0.3982	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-6.72	96.91		0.0693
20		82.80	-6.45	96.91		0.0666
30	63.59	165.60		110.33	0.3840	
40	63.94	165.60		110.33	0.3861	
50	64.20	165.60		110.33	0.3877	
60	64.42	165.60		110.33	0.3890	
70	64.64	165.60		110.33	0.3903	
80	64.86	165.60		110.33	0.3917	
90	65.12	165.60		110.33	0.3932	
100	65.34	165.60		110.33	0.3946	
110	65.56	165.60		110.33	0.3959	
120	65.82	165.60		110.33	0.3974	
130	65.93	165.60		110.33	0.3982	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.15	96.92		0.0738
20		82.80	-6.89	96.92		0.0711
30		165.60	-4.38	110.33		0.0397
40		165.60	-4.02	110.33		0.0365
50		165.60	-3.76	110.33		0.0341
60		165.60	-3.31	110.33		0.0300
70		165.60	-2.90	110.33		0.0263
80		165.60	-2.49	110.33		0.0226
90		165.60	-2.24	110.33		0.0203
100		165.60	-1.83	110.33		0.0166
110		165.60	-1.43	110.33		0.0129
120		165.60	-1.17	110.33		0.0106
130		165.60	-0.65	110.33		0.0059

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10		79.80	-6.96	80.76		0.0862
20		69.00	-6.71	80.76		0.0830
30	63.71	138.00		91.94	0.4617	
40	64.04	138.00		91.94	0.4641	
50	64.29	138.00		91.94	0.4659	
60	64.50	138.00		91.94	0.4674	
70	64.71	138.00		91.94	0.4689	
80	64.92	138.00		91.94	0.4705	
90	65.17	138.00		91.94	0.4723	
100	65.38	138.00		91.94	0.4738	
110	65.59	138.00		91.94	0.4753	
120	65.84	138.00		91.94	0.4771	
130	65.95	138.00		91.94	0.4779	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-6.96	80.76		0.0862
20		69.00	-6.71	80.76		0.0830
30		138.00	-4.25	91.94		0.0463
40		138.00	-3.92	91.94		0.0426
50		138.00	-3.67	91.94		0.0399
60		138.00	-3.23	91.94		0.0351
70		138.00	-2.83	91.94		0.0308
80		138.00	-2.43	91.94		0.0264
90		138.00	-2.18	91.94		0.0237
100		138.00	-1.79	91.94		0.0194
110		138.00	-1.39	91.94		0.0151
120		138.00	-1.14	91.94		0.0124
130		138.00	-0.63	91.94		0.0069

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.15	96.91		0.0738
20		82.80	-6.89	96.91		0.0711
30	63.59	165.60		110.33	0.3840	
40	63.94	165.60		110.33	0.3861	
50	64.20	165.60		110.33	0.3877	
60	64.42	165.60		110.33	0.3890	
70	64.64	165.60		110.33	0.3903	
80	64.86	165.60		110.33	0.3917	
90	65.12	165.60		110.33	0.3932	
100	65.34	165.60		110.33	0.3946	
110	65.56	165.60		110.33	0.3959	
120	65.82	165.60		110.33	0.3974	
130	65.93	165.60		110.33	0.3982	

Absolute Maximum of the all of the Stress Ratio's 0.4779

Governing Element: TOP HEAD

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
SKIRT-1	...	-5.8	1.1	...	...	+ Tens
SKIRT-1	...	-5.8	-1.1	...	-7.0	- Comp
SKIRT-2	...	-5.6	1.1	...	...	+ Tens
SKIRT-2	...	-5.6	-1.1	...	-6.7	- Comp
BOTTOM HEAD	66.4	-3.5	0.8	...	63.7	+ Tens
BOTTOM HEAD	66.4	-3.5	-0.8	...	...	- Comp



SHELL-1	66.4	-3.1	0.8	...	64.0	+ Tens
SHELL-1	66.4	-3.1	-0.8	...	...	- Comp
SHELL-2	66.4	-2.9	0.8	...	64.3	+ Tens
SHELL-2	66.4	-2.9	-0.8	...	...	- Comp
SHELL-3	66.4	-2.6	0.7	...	64.5	+ Tens
SHELL-3	66.4	-2.6	-0.7	...	...	- Comp
SHELL-4	66.4	-2.2	0.6	...	64.7	+ Tens
SHELL-4	66.4	-2.2	-0.6	...	...	- Comp
SHELL-5	66.4	-1.9	0.5	...	64.9	+ Tens
SHELL-5	66.4	-1.9	-0.5	...	...	- Comp
SHELL-6	66.4	-1.7	0.5	...	65.2	+ Tens
SHELL-6	66.4	-1.7	-0.5	...	...	- Comp
SHELL-7	66.4	-1.4	0.4	...	65.4	+ Tens
SHELL-7	66.4	-1.4	-0.4	...	...	- Comp
SHELL-8	66.4	-1.1	0.3	...	65.6	+ Tens
SHELL-8	66.4	-1.1	-0.3	...	...	- Comp
SHELL-9	66.4	-0.8	0.3	...	65.8	+ Tens
SHELL-9	66.4	-0.8	-0.3	...	...	- Comp
TOP HEAD	66.4	-0.5	0.1	...	66.0	+ Tens
TOP HEAD	66.4	-0.5	-0.1	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: SKIRT-2

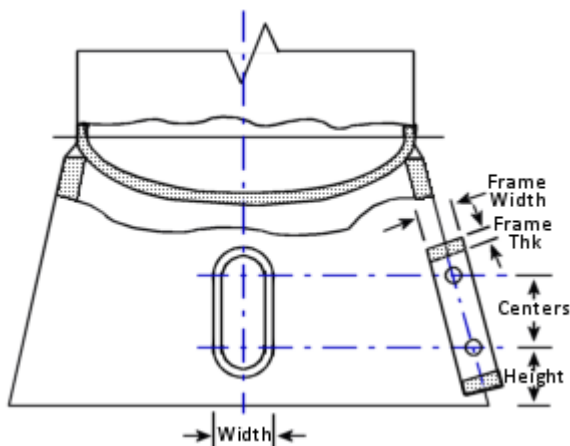
Total vessel (empty) weight	Fg	211742.031 Kgf
Head Weight below section 2-2	DeltaFg	20811.184 Kgf
Weight of vessel contents (liquid)	Ff	17566.201 Kgf
Vessel Gross Weight	Fg + Ff	229308.234 Kgf
Skirt Mean Diameter	Dz	6039.000 mm.
Skirt Analysis Thickness (Corroded)	ez	21.000 mm.
Skirt Height	H	1500.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	712958.312 Kg-m.
Applied Moment - Bottom of Skirt	Mb	770128.688 Kg-m.
Weight at Bottom of Skirt	W	233456.141 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	6033.600 mm.
Head Corroded Thickness	eb	30.400 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1501.6 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 6003.2 / (2 * 1501.6) = 1.9989 \\
 R &= D_i (0.44 * K + 0.02) = 6003.2 * (0.44 * 2.0 + 0.02) = 5400.0654 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 6003.2 * ((0.5 / 2.0) - 0.08) = 1021.3439 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	181624336.000 cm ⁴
Moment of Inertia	Iyy =	181624336.000 cm ⁴
Section Modulus	W =	601505152.000 mm. ³
Section Metal Area	A =	3984.137 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(0.18162\text{E}+09, 0.18162\text{E}+09) / 3984.137)} \\
 &= 2135.109 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 2135.109/1500.0$$

$$= 1.423$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 712958 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -712958/.24060E+10 - 233456.141/3984.13745$
 $= -8.652 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +712958/.24060E+10 - 233456.141/3984.13745$
 $= -2.841 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= 0.0 - 211742.031 - 17566.201 + 4 \cdot 712958/6041.0$
 $= 242762.1 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= 0.0 - 211742.031 - 17566.201 - 4 \cdot 712958/6041.0$
 $= -701378.5 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (242762.094 + 20811.184 + 17566.201) / (\pi \cdot 6033.6 \cdot 30.4) + 13.5 \cdot 6033.6 / (4 \cdot 30.4)$
 $= 71.77 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.7E+06 + 20811.184 + 17566.201) / (\pi \cdot 6033.6 \cdot 30.4) + 13.5 \cdot 6033.6 / (4 \cdot 30.4)$
 $= 55.71 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (17566.201 + 20811.184) / (\pi \cdot 6033.6 \cdot 30.4) + 13.5 \cdot 6033.6 / (4 \cdot 30.4)$
 $= 67.64 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= 242762.094 / (\pi \cdot 6041.0 \cdot 21.0)$
 $= 5.97 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -701378.500 / (\pi \cdot 6041.0 \cdot 21.0)$
 $= -17.26 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(6033.6 + 30.4 - 6039.0 + 21.0) / (1021.34 + 30.4)$
 $= 0.9791$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.979) = 11.74 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(30.4^2 + 21.0^2 + 2 \cdot 30.4 \cdot 21.0 \cdot 0.979)}$$

$$= 25.57 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 25.57 \cdot 242762.1 = 6207.49 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 25.57 \cdot -701378.5 = -17934.44 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{30.4}{21.0} \right)^2$$

$$= 0.511$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.511 \cdot 6 \cdot 6207.494 / (\pi \cdot 6033.6 \cdot 30.4^2)$$

$$= 10.65 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.511 \cdot 6 \cdot -17934.443 / (\pi \cdot 6033.6 \cdot 30.4^2)$$

$$= -30.76 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.511 \cdot 6 \cdot 6207.494 / (\pi \cdot 6041.0 \cdot 21.0^2)$$

$$= 22.28 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.511 \cdot 6 \cdot -17934.443 / (\pi \cdot 6041.0 \cdot 21.0^2)$$

$$= -64.37 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1.197/237.543 = 0.005$$

$$y = 125 \cdot eb/D_b = 125 \cdot 30.4/6033.6 = 0.6298$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.63 + 3.994 = 4.0996$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (13.5 + 0.0) \cdot 6033.6 \left(\left(\frac{11.74}{40.0} \right) \cdot 4.1 - 1 \right) / (4 \cdot 30.4)$$

$$= 13.61 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 71.773 - 10.645 + 13.611$$

$$= 74.74 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 71.773 + 10.645 - 13.611$$

$$= 68.81 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 55.705 - -30.756 + 13.611$$

$$= 100.07 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 55.705 + -30.756 - 13.611$$

$$= 11.34 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 67.642 + 10.645 + 13.611$$

$$= 91.90 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 67.642 - 10.645 - 13.611$$

$$= 43.39 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 67.642 + -30.756 + 13.611$$

$$= 50.50 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 67.642 - -30.756 - 13.611$$

$$= 84.79 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 5.974 - 22.281$$

$$= -16.31 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 5.974 + 22.281$$

$$= 28.25 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -17.259 - -64.373$$

$$= 47.11 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -17.259 + -64.373$$

$$= -81.63 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138.0(3 - (1 / 1.5)) \cdot (71.773 / 138.0)^2$$

$$= 389.11 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138.0(3 - (1 / 1.5)) \cdot (55.705 / 138.0)^2$$

$$= 399.01 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138.0(3 - (1 / 1.5)) \cdot (67.642 / 138.0)^2$$

$$= 391.90 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138.0(3 - (1 / 1.5)) \cdot (67.642 / 138.0)^2$$

$$= 391.90 \text{ N./mm}^2$$

For $\sigma_{3pi\text{tot}}$ & $\sigma_{3po\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (5.974/138.0)^2)$$

$$= 413.83 \text{ N./mm}^2$$

For $\sigma_{3qi\text{tot}}$ & $\sigma_{3qo\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-17.259/138.0)^2)$$

$$= 412.56 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	74.74	389.11	Pass	46
Stot1po	68.81	389.11	Pass	47
Stot1qi	100.07	399.01	Pass	48
Stot1qo	11.34	399.01	Pass	49
Stot2pi	91.90	391.90	Pass	50
Stot2po	43.39	391.90	Pass	51
Stot2qi	50.50	391.90	Pass	52
Stot2qo	84.79	391.90	Pass	53
Stot3pi	-16.31	413.83	Pass	54
Stot3po	28.25	413.83	Pass	55
Stot3qi	47.11	412.56	Pass	56
Stot3qo	-81.63	412.56	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	19771.309 mm.
Center of Gravity of the Liquid	3168.750 mm.
Center of Gravity of the Nozzles	12324.021 mm.
Center of Gravity of the Added Weights (Operating)	29568.219 mm.
Center of Gravity of the Added Weights (Empty)	29568.219 mm.
Center of Gravity of the Trunnions	28500.002 mm.
Center of Gravity of Bare Shell New and Cold	15133.121 mm.
Center of Gravity of Bare Shell Corroded	15092.226 mm.
Vessel CG in the Operating Condition	15616.168 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	15907.304 mm.
Vessel CG in the Test Condition	16387.018 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	29500.00 mm.
Total vessel weight (No Liquid) Twt	226654.89 Kg
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	339982.31 Kg
Elevation of the Tailing Lug (bottom)	0.00 mm.
Elevation of the Lifting Lug (top)	28500.00 mm.
Design Reaction force at the tailing lug	150220.83 Kg
Design Reaction force at the Trunnions	189761.48 Kg
CG distance during field lifting operation	15907.30 mm.
CG distance from tailing lug	15907.30 mm.
CG distance from the nearer lifting lug/trunnion	12592.69 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	10.07	13750.00	93.36 (UG-23)
Shear	2.61	0.00	79.80 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.4	0.0	111206.0	2.6
8950.00	822982.6	8.9	27196.5	0.4
16150.00	910509.8	9.8	39315.0	0.6
23350.01	522351.5	5.6	105361.6	1.7
29450.00	21147.3	0.2	4607.2	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.1079 at elevation 13749.9971 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

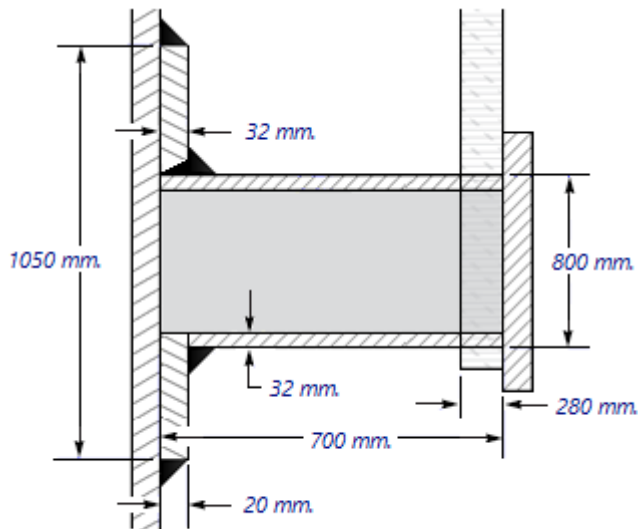
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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 800.000 mm.
Trunnion Thickness	Tthk 32.0000 mm.
Trunnion Length	Tl 700.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1050.000 mm.
Pad Thickness	Thkp 32.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 20.0000 mm.
Bail/Sling Width	Tt 280.0000 mm.
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Force and Stress Results for different Lift Angles (Kgf&N./mm²):

Bending Stress Allowable:	172.9 N./mm ²
Shear Stress Allowable:	104.8 N./mm ²
Bearing Stress Allowable:	196.5 N./mm ²

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	94880.7	36.6	12.1	4.2	0.326	Pass
5	95568.0	36.8	12.1	4.2	0.329	Pass
10	96253.0	37.1	12.2	4.2	0.331	Pass
15	96946.5	37.3	12.3	4.2	0.333	Pass
20	97659.4	37.6	12.4	4.3	0.336	Pass
25	98404.0	37.9	12.5	4.3	0.339	Pass
30	99194.8	38.2	12.6	4.3	0.341	Pass
35	100049.6	38.5	12.7	4.4	0.344	Pass
40	100991.5	38.9	12.8	4.4	0.347	Pass
45	102051.4	39.3	13.0	4.5	0.351	Pass
50	103272.7	39.8	13.1	4.5	0.355	Pass
55	104719.3	40.3	13.3	4.6	0.360	Pass
60	106489.3	41.0	13.5	4.7	0.366	Pass
65	108743.5	41.9	13.8	4.8	0.374	Pass
70	111765.1	43.1	14.2	4.9	0.384	Pass
75	116105.9	44.7	14.7	5.1	0.399	Pass
80	123005.1	47.4	15.6	5.4	0.423	Pass



85	135948.7	52.4	17.3	6.0	0.468	Pass
90	169991.2	65.5	21.6	7.4	0.585	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 113327.4 • 1.5
 = 169991.2 Kgf

Trunnion Section Modulus [Z]:
 = $\pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od})$
 = $\pi (800.0^4 - 736.0^4) / (32 \cdot 800.0)$
 = 14255651.0 mm.³

Trunnion Bending Moment [Tmom]:
 = $(T_l - 0.5 \cdot T_t) / 9.807 \cdot T_{lf}$
 = $(700.0 - 0.5 \cdot 280.0) / 9.807 \cdot 169991.16$
 = 95196.9 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 = Tmom • 9.807/Z
 = 95196.92 • 9.807/14255651
 = 65.49 N./mm² [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (172.92)

Trunnion Cross Section Area [Txsa]:
 = $\pi/4 (T_{od}^2 - T_{id}^2)$
 = $\pi/4 (800.0^2 - 736.0^2)$
 = 772.1 cm²

Trunnion Calculated Shear Stress [Ssc]:
 = Tlf/Txsa
 = 169991.16/772.08
 = 21.59 N./mm² [Passed] Ssc < 0.4 • Sy • Occfac (104.80)

Trunnion Unity Check [Uchk]:
 = Sbc/Sba + Ssc/Ssa
 = 65.49/172.92 + 21.59/104.8
 = 0.585 [Passed]

Weld Section Modulus at Edge of Trunnion [Sw]:
 = $\pi (T_{od}/2)^2$
 = $\pi (800.0/2)^2$
 = 502654.4 mm.²

Shear Stress in Weld due to Bending Moment [fs]:
 = Tmom/Sw
 = 95196.9/502654
 = 189.385 Kgf/mm.

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:
 = max(10mm, fs/(0.4 • Sy))
 = max(10mm, 189.385/(0.4 • 262.0))
 = 17.722 mm.

Trunnion Bearing Contact Area [Tbca]:
 = T_{od} • T_t
 = 800.0 • 280.0
 = 2240.0 cm²

Trunnion Bearing Stress [Sbec]:
 = Tlf/Tbca
 = 169991.16/2240.0
 = 7.44 N./mm² [Passed] Sbec < min(0.75 • SY • Occfac, 0.9 • Sy) (196.50)

Summary of Results:

Trunnion Unity Check Value: 0.5848 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	65.49	172.92	Pass
Shear Stress	21.59	104.80	Pass
Bearing Stress	7.44	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Loads on shell and WRC 107 results at different Lift Angles (Kgf & Kg-m.):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	94880.7	0.0	53134.3	0.0	0.505	1.138	Fail
5	95204.3	8329.3	53315.5	4664.5	0.507	1.142	Fail
10	94790.7	16714.2	53083.9	9360.1	0.505	1.137	Fail
15	93643.1	25091.6	52441.2	14051.6	0.499	1.123	Fail
20	91769.8	33401.5	51392.1	18705.2	0.489	1.101	Fail
25	89184.3	41587.3	49944.2	23289.4	0.475	1.070	Fail
30	85905.2	49597.4	48107.9	27775.1	0.458	1.031	Fail
35	81955.9	57386.1	45896.2	32136.9	0.438	0.984	Pass
40	77364.0	64916.1	43324.7	36353.7	0.414	0.930	Pass
45	72161.2	72161.2	40411.1	40411.1	0.387	0.868	Pass
50	66382.4	79111.5	37174.9	44303.3	0.357	0.800	Pass
55	60064.5	85781.0	33636.8	48038.3	0.329	0.725	Pass
60	53244.7	92222.5	29817.6	51645.6	0.351	0.645	Pass
65	45957.0	98555.1	25736.4	55191.9	0.374	0.559	Pass
70	38225.9	105024.8	21406.9	58815.1	0.397	0.589	Pass
75	30050.4	112149.7	16828.6	62805.1	0.423	0.627	Pass
80	21359.6	121136.4	11961.6	67837.7	0.456	0.676	Pass
85	11848.7	135431.3	6635.4	75843.0	0.509	0.755	Pass
90	0.0	169991.2	0.0	95196.9	0.638	0.947	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 5.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	6000.000 mm.
Vessel Thickness	Tv	32.000 mm.
Design Temperature	T1	37.8 C
Attachment Type	Type	Round
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	0.0000 mm.
Nozzle Diameter	Dn	800.000 mm.
Nozzle Thickness	Tn	32.000 mm.
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²

Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	32.000	mm.
Diameter of Reinforcing Pad	Dpad	1090.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust	No		

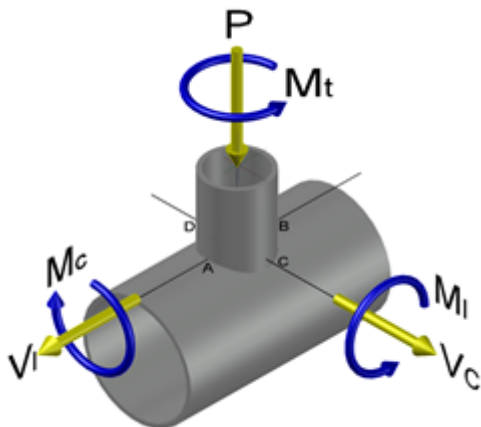
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	VL	8329.3	Kgf
Circumferential Shear (SUS)	Vc	95204.3	Kgf
Circumferential Moment (SUS)	Mc	53315.5	Kg-m.
Longitudinal Moment (SUS)	ML	4664.5	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 5.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 800.0 + 2 \cdot 1.65 \cdot \sqrt{3016.0 (32.0 - 0)}$$

$$= 1825.191 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	95204.3	Kgf
Longitudinal Shear	VL	8329.3	Kgf
Circumferential Moment	MC	53315.5	Kg-m.
Longitudinal Moment	ML	4664.5	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 47.38

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.115	4C	7.908	(A,B)
N(PHI) / (P/Rm)	0.115	3C	6.534	(C,D)
M(PHI) / (P)	0.115	2C1	0.068	(A,B)
M(PHI) / (P)	0.115	1C	0.102	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.115	3A	1.701	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.115	1A	0.090	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.115	3B	5.399	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.115	1B	0.040	(A,B,C,D)
N(x) / (P/Rm)	0.115	3C	6.534	(A,B)
N(x) / (P/Rm)	0.115	4C	7.908	(C,D)
M(x) / (P)	0.115	1C1	0.107	(A,B)
M(x) / (P)	0.115	2C	0.069	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.115	4A	2.542	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.115	2A	0.048	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.115	4B	1.615	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.115	2B	0.062	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-13.1	-13.1	13.1	13.1
Circ. Bend. MC		0.0	0.0	0.0	0.0	-196.7	196.7	196.7	-196.7
Circ. Memb. ML		-3.6	-3.6	3.6	3.6	0.0	0.0	0.0	0.0
Circ. Bend. ML		-7.6	7.6	7.6	-7.6	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-11.3	4.0	11.3	-4.0	-209.8	183.6	209.8	-183.6
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-19.6	-19.6	19.6	19.6
Long. Bend. MC		0.0	0.0	0.0	0.0	-104.3	104.3	104.3	-104.3
Long. Memb. ML		-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Long. Bend. ML		-11.9	11.9	11.9	-11.9	0.0	0.0	0.0	0.0
Tot. Long. Str.		-12.9	10.8	12.9	-10.8	-123.9	84.7	123.9	-84.7
Shear VC		11.6	11.6	-11.6	-11.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		11.6	11.6	-11.6	-11.6	-1.0	-1.0	1.0	1.0
Str. Int.		23.7	24.2	23.7	24.2	209.8	183.6	209.8	183.6

Dimensionless Parameters used : Gamma = 94.25

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.158	4C	12.137	(A,B)
N(PHI) / (P/Rm)	0.158	3C	6.994	(C,D)
M(PHI) / (P)	0.158	2C1	0.028	(A,B)
M(PHI) / (P)	0.158	1C	0.067	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.158	3A	3.664	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.158	1A	0.067	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.158	3B	8.648	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.158	1B	0.019	(A,B,C,D)
N(x) / (P/Rm)	0.158	3C	6.994	(A,B)
N(x) / (P/Rm)	0.158	4C	12.137	(C,D)



M(x) / (P)	0.158	1C1	0.058	(A,B)
M(x) / (P)	0.158	2C !	0.036	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.158	4A	7.853	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.158	2A	0.031	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.158	4B	3.618	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.158	2B	0.024	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-41.6	-41.6	41.6	41.6
Circ. Bend. MC		0.0	0.0	0.0	0.0	-431.0	431.0	431.0	-431.0
Circ. Memb. ML		-8.6	-8.6	8.6	8.6	0.0	0.0	0.0	0.0
Circ. Bend. ML		-10.6	10.6	10.6	-10.6	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-19.2	2.0	19.2	-2.0	-472.6	389.4	472.6	-389.4
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-89.2	-89.2	89.2	89.2
Long. Bend. MC		0.0	0.0	0.0	0.0	-197.9	197.9	197.9	-197.9
Long. Memb. ML		-3.6	-3.6	3.6	3.6	0.0	0.0	0.0	0.0
Long. Bend. ML		-13.6	13.6	13.6	-13.6	0.0	0.0	0.0	0.0
Tot. Long. Str.		-17.2	10.0	17.2	-10.0	-287.1	108.7	287.1	-108.7
Shear VC		17.0	17.0	-17.0	-17.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		17.0	17.0	-17.0	-17.0	-1.5	-1.5	1.5	1.5
Str. Int.		35.3	35.0	35.3	35.0	472.6	389.4	472.6	389.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-3.6	-3.6	3.6	3.6	-13.1	-13.1	13.1	13.1
Circ. Q (SUS)		-7.6	7.6	7.6	-7.6	-196.7	196.7	196.7	-196.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-1.1	-1.1	1.1	1.1	-19.6	-19.6	19.6	19.6
Long. Q (SUS)		-11.9	11.9	11.9	-11.9	-104.3	104.3	104.3	-104.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		11.6	11.6	-11.6	-11.6	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		23.4	23.4	23.4	23.4	19.7	19.7	19.7	19.7

Pm+Pl+Q (Total)	23.7	24.2	23.7	24.2	209.8	183.6	209.8	183.6
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	23.36	207.00	Passed
Pm+Pl+Q (TOTAL)	209.81	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-8.6	-8.6	8.6	8.6	-41.6	-41.6	41.6	41.6
Circ. Q (SUS)		-10.6	10.6	10.6	-10.6	-431.0	431.0	431.0	-431.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-3.6	-3.6	3.6	3.6	-89.2	-89.2	89.2	89.2
Long. Q (SUS)		-13.6	13.6	13.6	-13.6	-197.9	197.9	197.9	-197.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		17.0	17.0	-17.0	-17.0	-1.5	-1.5	1.5	1.5
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		34.4	34.4	34.4	34.4	89.3	89.3	89.3	89.3
Pm+Pl+Q (Total)		35.3	35.0	35.3	35.0	472.6	389.4	472.6	389.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	89.25	207.00	Passed
Pm+Pl+Q (TOTAL)	472.65	414.00	Failed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	6060.0000	mm.
Skirt Thickness	STHK	22.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	5838.5000	mm.
Outside Diameter of Basing	DOU, D5	6439.5000	mm.
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	21.5100	cm ²
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	172.00	N./mm ²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	6247.5000	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	20.0000	mm.
Width of Gussets at Top Plate	TWDT	182.0000	mm.
Width of Gussets at Base Plate	BWDT	182.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Gussets	HG, h1	312.0000	mm.
Distance between Gussets	RG, b6	120.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	50.0000	mm.
Radial Width of the Top Plate	TOPWITH, b5	192.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	226654.9	Kgf
Operating Weight of Vessel	ROW	244221.1	Kgf
Earthquake Moment on Basing	EQMOM	864703.5	Kg-m.
Wind Moment on Basing	WIMOM	232536.9	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 864704/6247.5) - 236562) / 24$$

$$= 13210.8 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 232537/6247.5) - 226655) / 24$$

$$= -3240.6 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/6247.5) - 244221) / 24$$

$$= -10175.9 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in one Bolt	Abss :	21.5100	cm ²
Area Available in all the Bolts	Abss * RN :	516.2400	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.274	0.410	1.437	2.500	0.443	0.780	42982.2	279544.5	1350.2
4	0.342	0.376	1.620	2.351	0.428	0.783	47316.9	283879.2	1580.5
6	0.393	0.385	1.748	2.239	0.417	0.784	50662.0	287224.3	1777.0
8	0.380	0.378	1.717	2.267	0.420	0.784	49817.5	286379.8	1725.7
10	0.379	0.379	1.714	2.269	0.420	0.784	49747.4	286309.7	1721.5
11	0.379	0.379	1.713	2.270	0.420	0.784	49712.4	286274.7	1719.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 49712.4 / (2.63 \cdot 6247.5 \cdot 2.27)$$

$$= 26.135 \text{ N./mm}^2, \text{ Should be } \leq 172.0 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 24 \cdot 21.51 / (\pi \cdot 6247.5)$$

$$= 2.63 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 1.621[(2 \cdot 0.3788 \cdot 6247.5 + 300.5) / (2 \cdot 0.3788 \cdot 6247.5)]$$

$$= 1.724 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 189.75, b = 408.898, l/b = 0.464052

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(724.896, 2167.9) = 2167.9 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 2167.9 / 172.9)} + 1.0$$

$$= 28.161 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 192.0, b = 120.0, b/l = 0.625

As b/l (0.625) is less than 1, inverting b/l = 1.6.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4 \cdot \pi)]$$

$$= 37725.27 / (4 \cdot \pi) [1.3(\ln((2 \cdot 192.0 \sin(\pi \cdot 93.75/192.0)) / (\pi \cdot 53.975))) + 1] - [(0.7 - 0.06) \cdot 37725.27 / (4 \cdot \pi)]$$

$$= 4267.3345 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{6 \cdot \text{Abs}(4267.33)/174.66} + 1.0$$

$$= 38.9155 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 37725.27 / (124.11 \cdot 182.0) + 1.0$$

$$= 17.379 \text{ (not less than } 9.525 + 1.0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 6247.5 / 24 = 817.796 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (37725 \cdot 93.75 / (817.8 \cdot 350.0 \cdot 171))^{2/3} \cdot 3030.0^{1/3} + \text{Ca}$$

$$= 20.243 + 1.0 = 21.243 \text{ mm.}$$

Shear Stress in a Single Bolt [rb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 38293.9 / (21.51 \cdot 24)$$

$$= 7.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	28.1607	mm.
Actual Basing Thickness as entered by user	38.0000	mm.
Required Thickness of Chair Cap	38.9155	mm.
Actual Top Ring Thickness as entered by user	50.0000	mm.
Required Gusset thickness, + CA	17.3788	mm.
Actual Gusset Thickness as entered by user	20.0000	mm.
Required Thickness of Skirt for Local Stress	21.2433	mm.
Given Thickness of Skirt	22.0000	mm.
Required Gusset Height to meet local stress	293.2531	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 27.952 \cdot (22.0 - 1.0)$$

$$= 59.86 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 60 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 4.511 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 37725.3 / (152.0 + 2 \cdot (312.0 + 50.0))$$

$$= 43.065 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 37725.3 \cdot 93.75 / (152.0 \cdot (362.0) + 0.6667 \cdot (362.0)^2)$$

$$= 24.839 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{43.07^2 + 24.84^2}$$

$$= 49.715 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 49.72 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 3.746 \text{ mm.}$$



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
 = Bolt Load / (2 • TopWth)
 = 37725.3 / (2 • 182.0)
 = 103.641 Kg/mm.

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 37725.3 • 93.75 / (2 • 362.0 • 182.0)
 = 26.841 Kg/mm.

Resultant Weld Load [Wr]:
 $= \sqrt{Wv^2 + Wh^2}$
 $= \sqrt{103.64^2 + 26.84^2}$
 = 107.06 Kg/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 107.06 / [(0.4 • 230) • 2 • 0.707]
 = 8.068 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	8.0679	mm.
Required Gusset to Top Plate Double Fillet Weld Size	8.0679	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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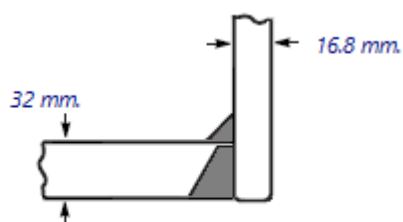
Input, Nozzle Desc: N2-2"
From: 30

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3862.0	mm.
Centerline Distance to Tangent		2140.0	mm.

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$
 $= 26.4411$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2654$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.1280	mm.
Parallel to Vessel Wall	Rn+tn+t	72.5640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.9100	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

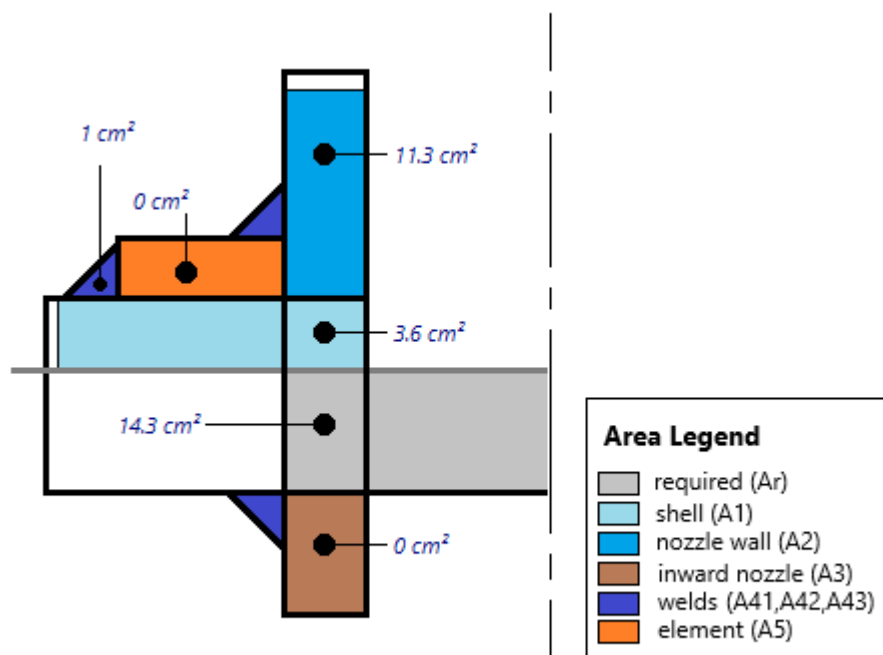


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	14.278	NA	NA
Area in Shell	A1	3.608	NA	NA
Area in Nozzle Wall	A2	11.296	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.904	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (54.0 \cdot 26.4411 \cdot 1.0 + 2 \cdot 15.164 \cdot 26.4411 \cdot 1.0 \cdot (1 - 1.))$
 $= 14.278 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 91.128(1.0 \cdot 30.4 - 1.0 \cdot 26.441) - 2 \cdot 15.164$
 $(1.0 \cdot 30.4 - 1.0 \cdot 26.441) \cdot (1 - 1.)$
 $= 3.608 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 37.91)(15.16 - 0.27) \cdot 1.$
 $= 11.296 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.8654 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 30.9731 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 30.9731 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 6.4000 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(30.9731, 3.1)]$
 $= 6.4000 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8654, 6.4)$
 $= 6.4000 \text{ mm}.$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $tg = 16.764$, $tr = 0.265$, $c = 1.6 \text{ mm}.$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 13.50/51.00 = 0.265

Weld Size Calculations, Description: N2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.1640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (14.2783 - 3.6077 + 2 \cdot 15.164 \cdot 1. \cdot (1.0 \cdot 30.4 - 26.4411)) \cdot 138)$
 $= 16705.09 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (11.2959 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 17302.52 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (11.2959 + 0.0 + 1. + (9.2195)) \cdot 138$
 $= 30276.10 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (11.2959 + 0.0 + 1. + 0.0 + (9.2195)) \cdot 138$
 $= 30276.10 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 9133. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 34.582) \cdot (16.764 - 1.6) \cdot 0.7 \cdot 138$
 $= 16228. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 84.328 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 41932. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9133 + 16228) = 25361 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9133 + 0 + 41932 + 0) = 51066 \text{ Kgf}$

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (9133 + 41932 + 0) = 51066 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 25360 Kgf, must exceed W = 16705 Kgf or W1 = 17302 Kgf
Path 2-2 = 51065 Kgf, must exceed W = 16705 Kgf or W2 = 30276 Kgf
Path 3-3 = 51065 Kgf, must exceed W = 16705 Kgf or W3 = 30276 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.1636 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 192.1635 mm.

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Input, Nozzle Desc: N6-2"

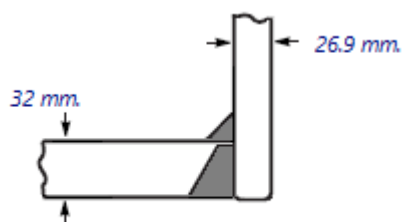
From: 40

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2654 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 165.4480 mm.
 Parallel to Vessel Wall Rn+tn+t 82.7240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 63.3100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

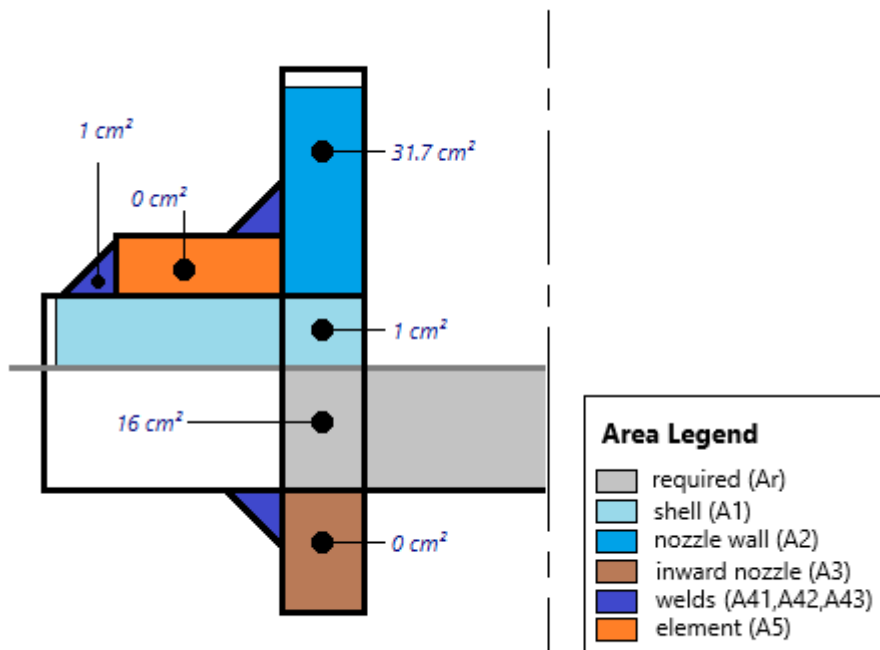


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.951	NA	NA
Area in Shell	A1	0.960	NA	NA
Area in Nozzle Wall	A2	31.729	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	33.689	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (54.0 \cdot 29.5386 \cdot 1.0 + 2 \cdot 25.324 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 15.951 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 111.448(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 25.324$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 0.960 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 63.31)(25.32 - 0.27) \cdot 1.$
 $= 31.729 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.8654 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 31.1386 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 31.1386 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(31.1386, 3.1)]$
 $= 6.8578 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8654, 6.8578)$
 $= 6.8578 \text{ mm}.$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $tg = 26.924$, $tr = 0.265$, $c = 1.6 \text{ mm}.$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.01$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A 21 $^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) -104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 $^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50/153.10 = 0.088$

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (15.9511 - 0.96 + 2 \cdot 25.324 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 21709.10 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (31.7286 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 46055.05 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (31.7286 + 0.0 + 1. + (15.3967)) \cdot 138$
 $= 67721.03 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (31.7286 + 0.0 + 1. + 0.0 + (15.3967)) \cdot 138$
 $= 67721.03 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 104.648 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 39.662) \cdot (26.924 - 1.6) \cdot 0.7 \cdot 138$
 $= 31081. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 104.648 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 52036. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 31081) = 42415 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 52036 + 0) = 63370 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 52036 + 0) = 63370 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 42415 Kgf, must exceed W = 21709 Kgf or W1 = 46055 Kgf
Path 2-2 = 63370 Kgf, must exceed W = 21709 Kgf or W2 = 67721 Kgf
Path 3-3 = 63370 Kgf, must exceed W = 21709 Kgf or W3 = 67721 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.4562 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.7812 mm.

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Input, Nozzle Desc: K1B-3"

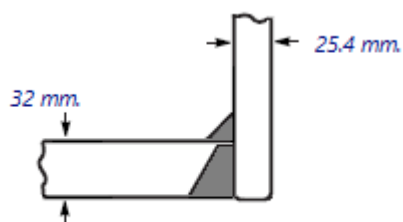
From: 40

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.4000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	260.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K1B-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	1.000	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 39.7 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.3903 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.8000 mm.
 Parallel to Vessel Wall Rn+tn+t 93.9000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

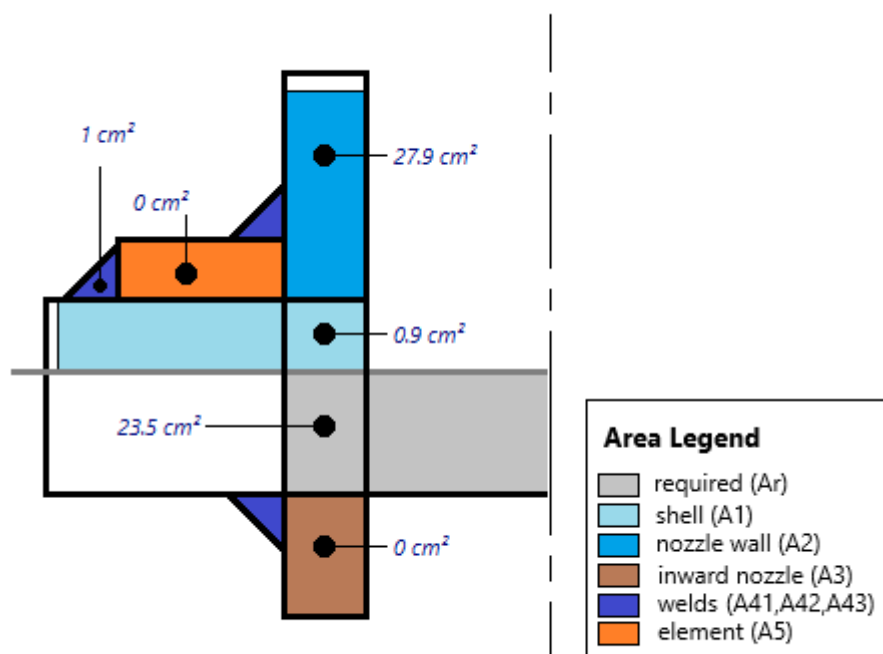


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	23.454	NA	NA
Area in Shell	A1	0.934	NA	NA
Area in Nozzle Wall	A2	27.857	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.791	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.4 \cdot 29.5386 \cdot 1.0 + 2 \cdot 23.8 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 23.454 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 108.4(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 23.8$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 0.934 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.5)(23.8 - 0.39) \cdot 1.$
 $= 27.857 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.9903 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 31.1386 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 31.1386 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 7.3300 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.33, \max(31.1386, 3.1)]$
 $= 7.3300 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9903, 7.33)$
 $= 7.3300 \text{ mm}.$

Available Nozzle Neck Thickness = 25.4000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, $tg = 25.4$, $tr = 0.39$, $c = 1.6 \text{ mm}.$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50/153.10 = 0.088$

Weld Size Calculations, Description: K1B-3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (23.4539 - 0.9338 + 2 \cdot 23.8 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 32266.88 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27.857 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 40607.08 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27.857 + 0.0 + 1. + (14.4701)) \cdot 138$
 $= 60969.20 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27.857 + 0.0 + 1. + 0.0 + (14.4701)) \cdot 138$
 $= 60969.20 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 127.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 13755. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 51.6) \cdot (25.4 - 1.6) \cdot 0.7 \cdot 138$
 $= 38003. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 127.0 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 63151. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (13755 + 38003) = 51758 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (13755 + 0 + 63151 + 0) = 76906 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (13755 + 63151 + 0) = 76906 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 51757 Kgf, must exceed W = 32266 Kgf or W1 = 40607 Kgf
Path 2-2 = 76905 Kgf, must exceed W = 32266 Kgf or W2 = 60969 Kgf
Path 3-3 = 76905 Kgf, must exceed W = 32266 Kgf or W3 = 60969 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.6721 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.6720 mm.

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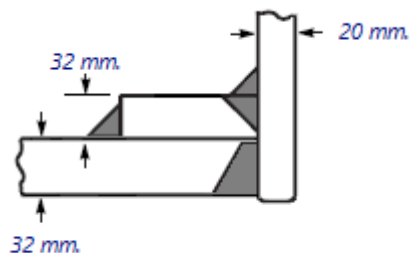
Input, Nozzle Desc: M1-24"
From: 50

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1150.0000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	30.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		250.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 306.4 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 3.0153 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.6000 mm.
Parallel to Vessel Wall, opening length	d	612.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		76.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

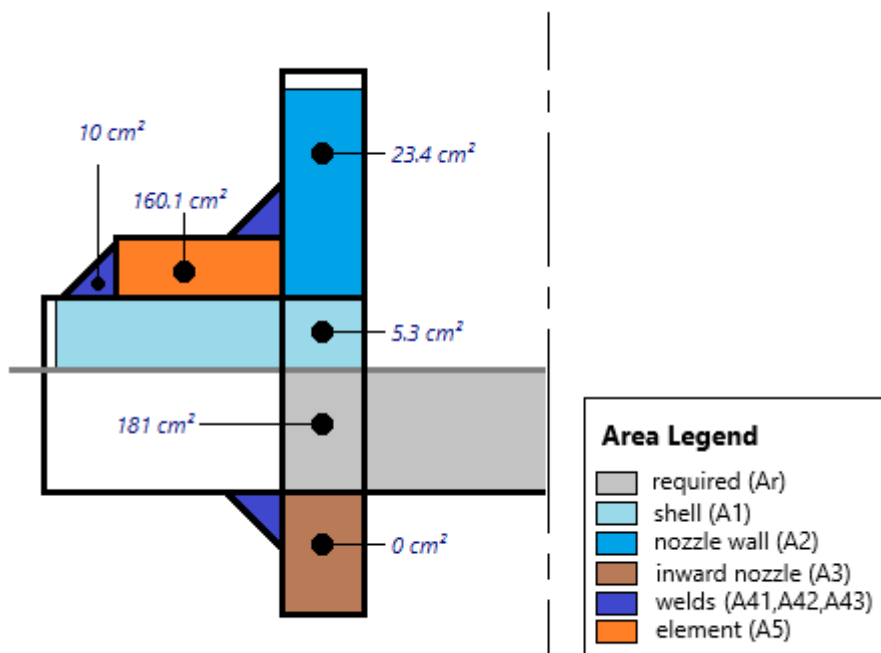


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	181.013	NA	NA
Area in Shell	A1	5.279	NA	NA
Area in Nozzle Wall	A2	23.384	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	10.000	NA	NA
Area in Element	A5	160.125	NA	NA
TOTAL AREA AVAILABLE	Atot	198.788	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1094.4506 32.0000 mm.
 Based on given Pad Diameter: 1150.0000 28.4477 mm.
 Based on the Estimated Diameter Limit: 1224.0125 24.7822 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.8 \cdot 29.5386 \cdot 1.0 + 2 \cdot 18.4 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 181.013 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.8(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 18.4$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 5.279 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 76.0) \cdot (18.4 - 3.02) \cdot 1.$

= 23.384 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 30.0^2 \cdot 1.$
 = 10.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1150.0 - 649.6) \cdot 32.0 \cdot 1.$
 = 160.125 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.6153 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 4.6153 mm.

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 32.0$, $t_r = 29.539$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.972$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	4 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: 4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 13.50/19.60 = 0.689

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 18.4000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	21.2100 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (181.0126 - 5.2788 + 2 \cdot 18.4 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 247735.53 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (23.3843 + 160.1251 + 10.0 - 0.0 \cdot 1.) \cdot 138$
 $= 272302.78 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (23.3843 + 0.0 + 1. + (11.187)) \cdot 138$
 $= 50055.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (23.3843 + 0.0 + 10.0 + 160.1251 + (11.187)) \cdot 138$
 $= 288044.94 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 649.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 70357. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1150.0 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 373661. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 315.6) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 179699. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 649.6 \cdot 32.0 \cdot 0.74 \cdot 138$
 $= 340009. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 649.6 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 323014. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (373661 + 179699) = 553360 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (70357 + 340009 + 323014 + 0) = 733380 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (373661 + 323014 + 0) = 696675 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 553359 Kgf, must exceed W = 247735 Kgf or W1 = 272302 Kgf
Path 2-2 = 733379 Kgf, must exceed W = 247735 Kgf or W2 = 50055 Kgf
Path 3-3 = 696675 Kgf, must exceed W = 247735 Kgf or W3 = 288044 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 17.6344 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.6344 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

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Input, Nozzle Desc: M2-24"

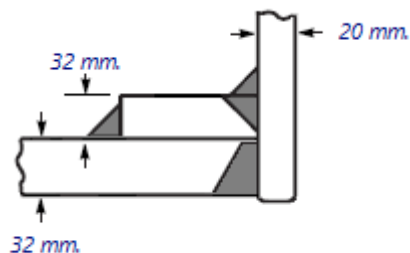
From: 50

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1150.0000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	30.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		250.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: M2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 306.4 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 3.0153 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.6000 mm.
Parallel to Vessel Wall, opening length	d	612.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		76.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

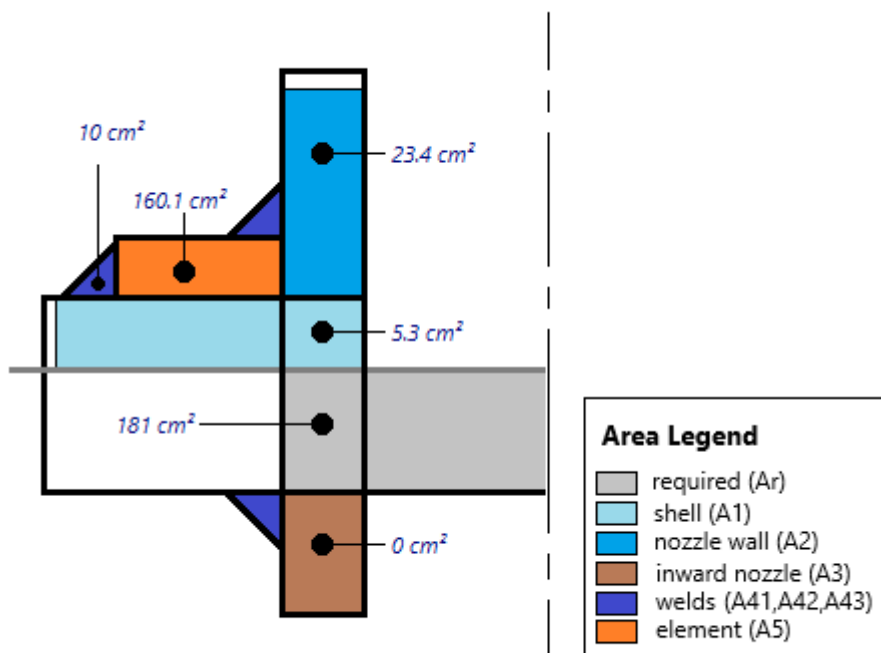


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	181.013	NA	NA
Area in Shell	A1	5.279	NA	NA
Area in Nozzle Wall	A2	23.384	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	10.000	NA	NA
Area in Element	A5	160.125	NA	NA
TOTAL AREA AVAILABLE	Atot	198.788	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1094.4506 32.0000 mm.
 Based on given Pad Diameter: 1150.0000 28.4477 mm.
 Based on the Estimated Diameter Limit: 1224.0125 24.7822 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.8 \cdot 29.5386 \cdot 1.0 + 2 \cdot 18.4 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 181.013 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.8(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 18.4$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 5.279 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 76.0) \cdot (18.4 - 3.02) \cdot 1.$

= 23.384 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 30.0^2 \cdot 1.$
 = 10.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1150.0 - 649.6) \cdot 32.0 \cdot 1.$
 = 160.125 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.6153 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 4.6153 mm.

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 32.0$, $t_r = 29.539$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.972$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 20.0$, $t_r = 3.015$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.164$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 13.50/19.60 = 0.689

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 18.4000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	21.2100 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (181.0126 - 5.2788 + 2 \cdot 18.4 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 247735.53 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (23.3843 + 160.1251 + 10.0 - 0.0 \cdot 1.) \cdot 138$
 $= 272302.78 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (23.3843 + 0.0 + 1. + (11.187)) \cdot 138$
 $= 50055.26 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (23.3843 + 0.0 + 10.0 + 160.1251 + (11.187)) \cdot 138$
 $= 288044.94 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 649.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 70357. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1150.0 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 373661. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 315.6) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 179699. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 649.6 \cdot 32.0 \cdot 0.74 \cdot 138$
 $= 340009. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 649.6 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 323014. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (373661 + 179699) = 553360 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (70357 + 340009 + 323014 + 0) = 733380 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (373661 + 323014 + 0) = 696675 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 553359 Kgf, must exceed W = 247735 Kgf or W1 = 272302 Kgf

Path 2-2 = 733379 Kgf, must exceed W = 247735 Kgf or W2 = 50055 Kgf

Path 3-3 = 696675 Kgf, must exceed W = 247735 Kgf or W3 = 288044 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 17.6344 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.6344 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 261.93 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 612.80 mm.

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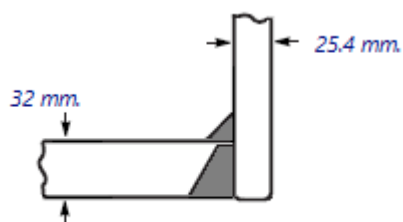
Input, Nozzle Desc: K1A-3"
From: 110

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		22550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.4000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	260.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.


Reinforcement CALCULATION, Description: K1A-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	1.000	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 39.7 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.3903$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.8000 mm.
 Parallel to Vessel Wall Rn+tn+t 93.9000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

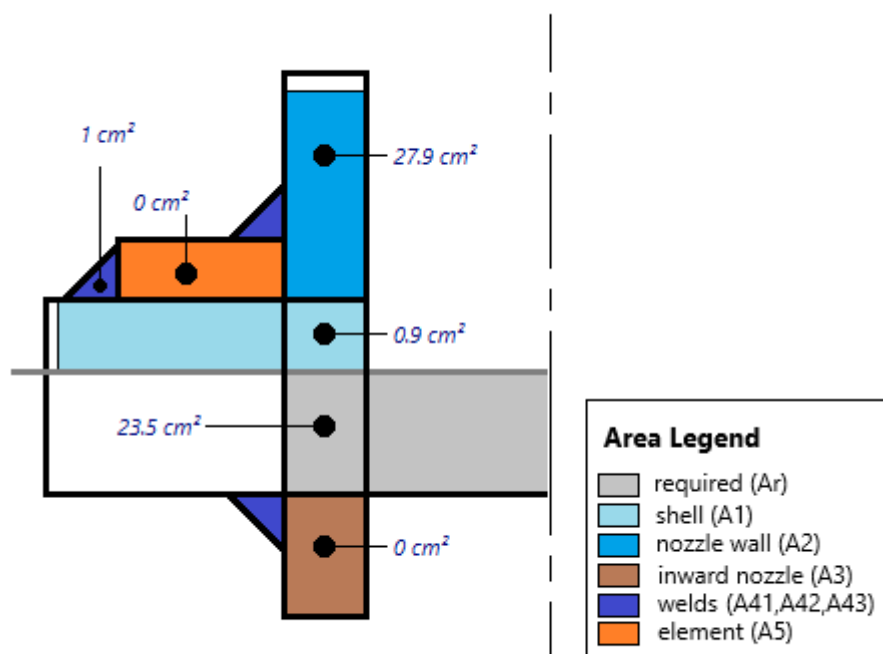


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	23.454	NA	NA
Area in Shell	A1	0.934	NA	NA
Area in Nozzle Wall	A2	27.857	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.791	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.4 \cdot 29.5386 \cdot 1.0 + 2 \cdot 23.8 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 23.454 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 108.4(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 23.8$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 0.934 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.5)(23.8 - 0.39) \cdot 1.$
 $= 27.857 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.9903 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 31.1386 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 31.1386 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 7.3300 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.33, \max(31.1386, 3.1)]$
 $= 7.3300 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9903, 7.33)$
 $= 7.3300 \text{ mm}.$

Available Nozzle Neck Thickness = 25.4000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 25.4$, $tr = 0.39$, $c = 1.6 \text{ mm}.$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(t_g - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50/153.10 = 0.088$

Weld Size Calculations, Description: K1A-3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot tr)) \cdot Sv)$
 $= \max(0, (23.4539 - 0.9338 + 2 \cdot 23.8 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 32266.88 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot f_r2) \cdot Sv$
 $= (27.857 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 40607.08 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot Sv$
 $= (27.857 + 0.0 + 1. + (14.4701)) \cdot 138$
 $= 60969.20 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27.857 + 0.0 + 1. + 0.0 + (14.4701)) \cdot 138$
 $= 60969.20 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 127.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 13755. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 51.6) \cdot (25.4 - 1.6) \cdot 0.7 \cdot 138$
 $= 38003. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 127.0 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 63151. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (13755 + 38003) = 51758 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (13755 + 0 + 63151 + 0) = 76906 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (13755 + 63151 + 0) = 76906 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 51757 Kgf, must exceed W = 32266 Kgf or W1 = 40607 Kgf
Path 2-2 = 76905 Kgf, must exceed W = 32266 Kgf or W2 = 60969 Kgf
Path 3-3 = 76905 Kgf, must exceed W = 32266 Kgf or W3 = 60969 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.6721 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.6720 mm.

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Input, Nozzle Desc: N1-10"

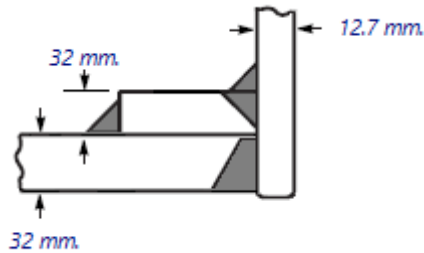
From: 120

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		25000.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		320.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	260.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	450.0000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	30.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		88.4750	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N1-10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 13.5 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 13.5)$
 $= 1.5549 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	501.7000 mm.
Parallel to Vessel Wall, opening length	d	250.8500 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		59.7500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.)$
 $= 0.855$

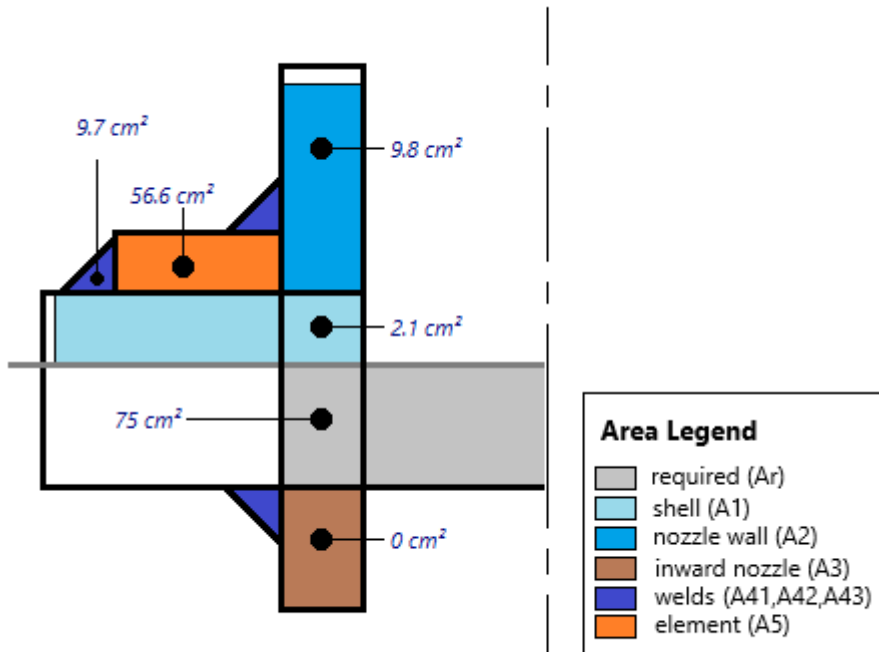


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	75.048	NA	NA
Area in Shell	A1	2.133	NA	NA
Area in Nozzle Wall	A2	9.753	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	9.683	NA	NA
Area in Element	A5	56.623	NA	NA
TOTAL AREA AVAILABLE	Atot	78.192	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 440.1750 32.0000 mm.
 Based on given Pad Diameter: 450.0000 30.2232 mm.
 Based on the Estimated Diameter Limit: 500.1125 23.5530 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (250.85 \cdot 29.5386 \cdot 1.0 + 2 \cdot 11.1 \cdot 29.5386 \cdot 1.0 \cdot (1 - 0.86))$
 $= 75.048 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 250.85(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 11.1$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 0.855)$
 $= 2.133 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 59.75) \cdot (11.1 - 1.55) \cdot 0.8551$

$$= 9.753 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + \text{Trapfr}_4$
 $= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 882.7774^2 \cdot 1.$
 $= 9.683 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 $= (450.0 - 273.05) \cdot 32.0 \cdot 1.$
 $= 56.623 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.1549 \text{ mm}.$
 Wall Thickness per UG16(b), $tr_{16b} = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb_1 = 31.1386 \text{ mm}.$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 31.1386 \text{ mm}.$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb_3 = 9.7026 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[9.703, \max(31.1386, 3.1)]$
 $= 9.7026 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.1549, 9.7026)$
 $= 9.7026 \text{ mm}.$

Available Nozzle Neck Thickness $= 0.875 \cdot 12.700 = 11.113 \text{ mm}.$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.9,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 264.1 N./mm ²	Passed
Occasional	: 7.3,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 15.4,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $tr = 1.555$, $c = 1.6 \text{ mm}.$, $E^* = 1.0$
 Thickness Ratio $= tr \cdot E^* / (t_g - c) = 0.163$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 1.555$, $c = 1.6 \text{ mm}.$, $E^* = 1.0$
 Thickness Ratio $= tr \cdot E^* / (t_g - c) = 0.163$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 1.555$, $c = 1.6 \text{ mm}.$, $E^* = 1.0$
 Thickness Ratio $= tr \cdot E^* / (t_g - c) = 0.163$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 32.0, tr = 29.539, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.972$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 11.113, tr = 1.555, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.163$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50 / 19.60 = 0.689$

Weld Size Calculations, Description: N1-10"

Intermediate Calc. for nozzle/shell Welds Tmin 11.1000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	21.2100 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (75.048 - 2.1332 + 2 \cdot 11.1 \cdot 0.8551 \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 102834.55 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / 707)^2 \cdot fr2) \cdot Sv$
 $= (9.7531 + 56.623 + 9.6827 - 0.0 \cdot 0.86) \cdot 138$
 $= 107028.52 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (9.7531 + 0.0 + 0.8551 + (5.7706)) \cdot 138$
 $= 23047.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (9.7531 + 0.0 + 9.6827 + 56.623 + (5.7706)) \cdot 138$
 $= 115148.81 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 25287. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.1416/2.0) \cdot 450.0 \cdot 30.0 \cdot 0.49 \cdot 138 \\
 &= 146215. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 130.975) \cdot (12.7 - 1.6) \cdot 0.7 \cdot 118 \\
 &= 38468. \text{ Kg}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 273.05 \cdot 32.0 \cdot 0.74 \cdot 138 \\
 &= 142918. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 273.05 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138 \\
 &= 135774. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (146215 + 38468) = 184684 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (25287 + 142918 + 135774 + 0) = 303980 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (146215 + 135774 + 0) = 281990 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 184683 Kg, must exceed W = 102834 Kg or W1 = 107028 Kg
 Path 2-2 = 303979 Kg, must exceed W = 102834 Kg or W2 = 23047 Kg
 Path 3-3 = 281989 Kg, must exceed W = 102834 Kg or W3 = 115148 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.1081 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 295.1081 mm.

Input Echo, WRC297 Item 1, Description: N1-10" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6000.000	mm.
Shell Thickness	32.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	273.050	mm.
Nozzle Thickness	12.7000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 450.000 mm.
 Thickness of Reinforcing Pad 32.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	13.50	bars
Radial Load	P	1911.36	Kgf
Circumferential Shear	Vc	1274.24	Kgf
Longitudinal Shear	Vl	1274.24	Kgf
Circumferential Moment	Mc	892.01	Kg-m.
Longitudinal Moment	Ml	892.01	Kg-m.
Torsional Moment	Mt	1274.31	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(3032.8(32.0 - 1.6))}$
 $= 1275.061 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6065.600 mm.
Nozzle Outside Diameter	(d) =	273.050 mm.
Vessel Thickness used	(T) =	62.400 mm.
Nozzle Thickness used	(t) =	11.100 mm.
T / t	=	5.622
d / t	=	24.599

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.444$$

Nr/P	=	0.100
Mr/P	=	0.073
M0/P	=	0.216
N0/P	=	0.357
MrD/Mc	=	0.095
NrDL/Mc	=	0.109
M0d/Mc	=	0.311
N0DL/Mc	=	-0.109
MrD/Ml	=	0.096
NrDL/Ml	=	0.109
M0D/Ml	=	0.292
N0DL/Ml	=	0.157

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	64	65	64	65
Outplane Stress Summary	40	83	72	56



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	32	32	32	32
Inplane Stress Summary	23	37	35	29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	40	83	72	57

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	32	32	32	32
Outplane Stress Summary	9	52	38	20

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	64	65	64	65
Inplane Stress Summary	56	71	67	63

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0

Torsional Shear (Mt)	1	1	1	1
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Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	56	71	67	63

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19
Hoop Stress Summary	14	16	17	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-21	21	-21	21
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-64	64	64	-64
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-94	77	64	-21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	12	12	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	111	79	65	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19

Hoop Stress Summary 16 18 14 16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-21	21	-21	21
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-62	62	62	-62
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-92	75	63	-19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	76	66	43

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(3016.8(32.0 - 1.6))}$
 $= 1272.415 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6033.600 mm.
 Nozzle Outside Diameter (d) = 450.000 mm.
 Vessel Thickness used (T) = 30.400 mm.
 Nozzle Thickness used (t) = 99.575 mm.
 T / t = 0.305
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.051$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.103
 Mr/P = 0.220
 M0/P = 0.064
 N0/P = 0.120
 MrD/Mc = 0.309
 NrDL/Mc = 0.078
 M0d/Mc = 0.095
 N0DL/Mc = 0.046
 MrD/Ml = 0.188
 NrDL/Ml = 0.093
 M0D/Ml = 0.059
 N0DL/Ml = 0.113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	132	133	132	133
Outplane Stress Summary	112	144	132	134

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	11	112	63	69

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	112	144	132	134

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-12	12	12	-12
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66
Outplane Stress Summary	43	82	69	60

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-39	39	39	-39
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	132	133	132	133

Inplane Stress Summary	63	196	144	121
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	63	196	144	121
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	64	65	64	65	64	65	64	65
Circ. Pl (SUS)	-2	-2	0	0	0	0	0	0
Circ. Q (SUS)	-20	20	8	-8	-6	6	2	-2
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	0	0	0	0	-1	-1	-1	-1
Long. Q (SUS)	-6	6	2	-2	-21	21	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	64.0	65.0	64.0	65.0	64.0	65.0	64.0	65.0
Pm+Pl (SUS)	62.0	63.0	64.0	65.0	64.0	65.0	64.0	65.0
Pm+Pl+Q (Total)	42.1	83.0	72.0	57.0	58.0	71.1	66.0	63.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	83.02	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	133	132	133	132	133	132	133



Circ. P1 (SUS)	-4	-4	0	0	-3	-3	-1	-1
Circ. Q (SUS)	-14	14	0	0	-65	65	13	-13
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. P1 (SUS)	-3	-3	-1	-1	-2	-2	-2	-2
Long. Q (SUS)	-49	49	-3	3	-19	19	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	132.0	133.0	132.0	133.0	132.0	133.0	132.0	133.0
Pm+P1 (SUS)	128.0	129.0	132.0	133.0	129.0	130.0	131.0	132.0
Pm+P1+Q (Total)	114.0	143.0	132.0	133.0	64.1	195.0	144.0	119.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+P1 (SUS)	133.00	207.00	Passed
Pm+P1+Q (TOTAL)	195.01	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	19	17	19	17	19	17	19
Circ. P1 (SUS)	-2	-2	0	0	-1	-1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. P1 (SUS)	-17	-17	13	13	-17	-17	13	13
Long. Q (SUS)	-85	85	43	-43	-83	83	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	17.0	19.0	17.0	19.0	17.0	19.0	17.0	19.0
Pm+P1 (SUS)	24.3	26.3	21.8	22.2	25.3	27.3	21.7	22.0
Pm+P1+Q (Total)	111.2	78.0	65.0	43.3	108.9	74.9	64.5	43.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	118.00	Passed
Pm+P1 (SUS)	27.29	177.00	Passed
Pm+P1+Q (TOTAL)	111.20	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

Input, Nozzle Desc: K2-2"

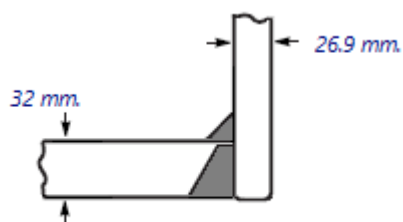
From: 120

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		25000.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		50.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	260.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13.5 \cdot 3001.6001 / (138 \cdot 1.0 - 0.6 \cdot 13.5)$
 $= 29.5386 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2654 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 165.4480 mm.
 Parallel to Vessel Wall Rn+tn+t 82.7240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 63.3100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

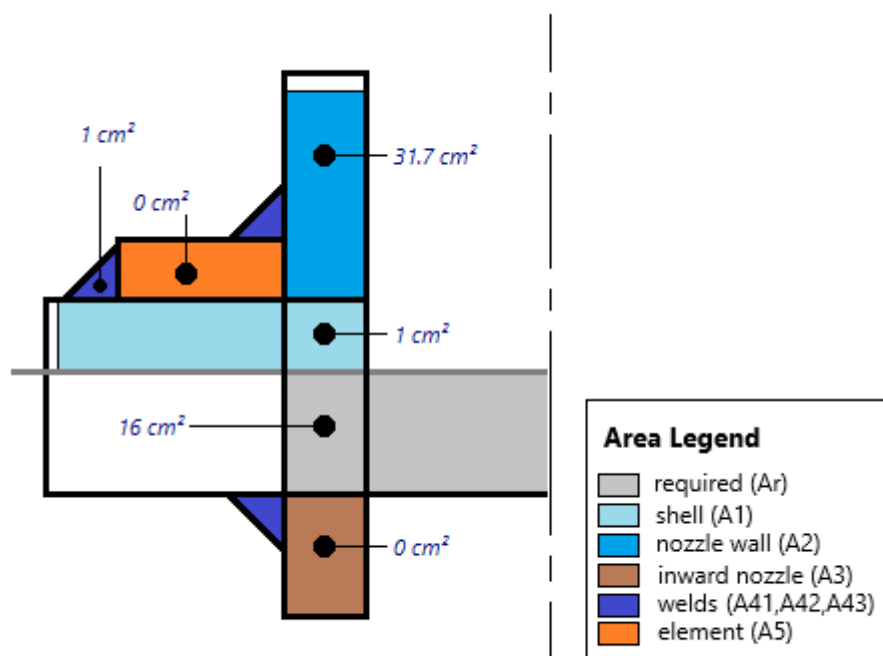


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.951	NA	NA
Area in Shell	A1	0.960	NA	NA
Area in Nozzle Wall	A2	31.729	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	33.689	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (54.0 \cdot 29.5386 \cdot 1.0 + 2 \cdot 25.324 \cdot 29.5386 \cdot 1.0 \cdot (1 - 1.))$
 $= 15.951 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 111.448(1.0 \cdot 30.4 - 1.0 \cdot 29.539) - 2 \cdot 25.324$
 $(1.0 \cdot 30.4 - 1.0 \cdot 29.5386) \cdot (1 - 1.)$
 $= 0.960 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 63.31)(25.32 - 0.27) \cdot 1.$
 $= 31.729 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.8654 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 31.1386 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 31.1386 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(31.1386, 3.1)]$
 $= 6.8578 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8654, 6.8578)$
 $= 6.8578 \text{ mm}.$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $tg = 26.924$, $tr = 0.265$, $c = 1.6 \text{ mm}.$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.01$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A 21 $^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) -104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 $^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50/153.10 = 0.088$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$
 $= \max(0, (15.9511 - 0.96 + 2 \cdot 25.324 \cdot 1. \cdot (1.0 \cdot 30.4 - 29.5386)) \cdot 138)$
 $= 21709.10 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (31.7286 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 46055.05 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (31.7286 + 0.0 + 1. + (15.3967)) \cdot 138$
 $= 67721.03 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (31.7286 + 0.0 + 1. + 0.0 + (15.3967)) \cdot 138$
 $= 67721.03 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 104.648 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 39.662) \cdot (26.924 - 1.6) \cdot 0.7 \cdot 138$
 $= 31081. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 104.648 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 52036. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 31081) = 42415 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 52036 + 0) = 63370 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 52036 + 0) = 63370 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 42415 Kgf, must exceed W = 21709 Kgf or W1 = 46055 Kgf
Path 2-2 = 63370 Kgf, must exceed W = 21709 Kgf or W2 = 67721 Kgf
Path 3-3 = 63370 Kgf, must exceed W = 21709 Kgf or W3 = 67721 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.4562 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.4561 mm.

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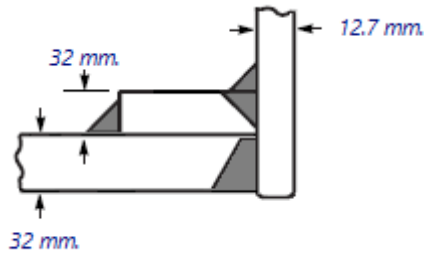
Input, Nozzle Desc: N3-8"
From: 130

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	450.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	550.0000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	30.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		165.4625	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N3-8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$
 $= 26.4411 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 13.5 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 13.5)$
 $= 1.2476 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	393.7500 mm.
Parallel to Vessel Wall, opening length	d	196.8750 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		59.7500 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.)$
 $= 0.855$

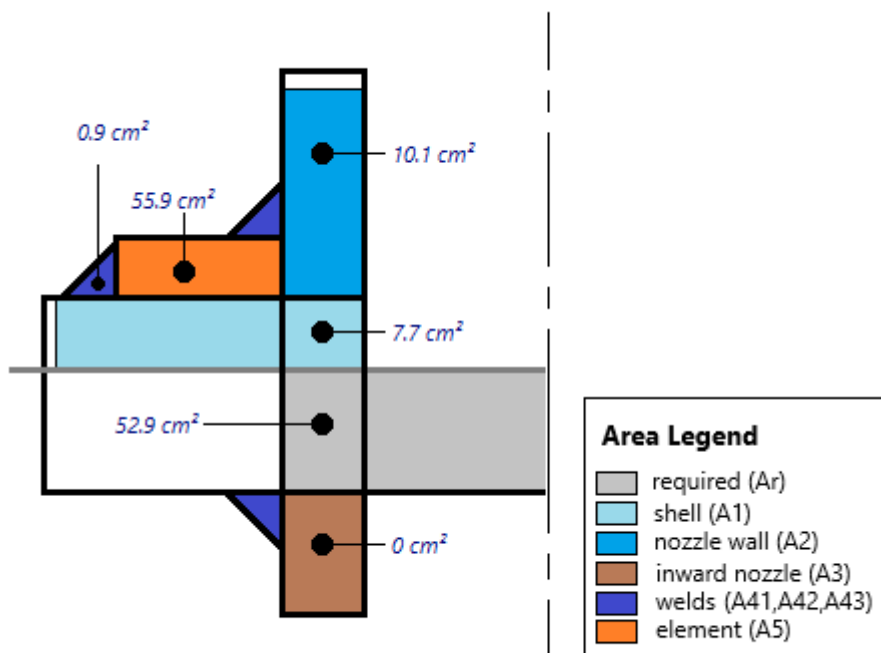


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	52.907	NA	NA
Area in Shell	A1	7.667	NA	NA
Area in Nozzle Wall	A2	10.067	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	55.895	NA	NA
TOTAL AREA AVAILABLE	Atot	74.484	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 326.3199 32.0000 mm.
 Based on given Pad Diameter: 550.0000 19.6470 mm.
 Based on the Estimated Diameter Limit: 392.1625 19.8272 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (196.875 \cdot 26.4411 \cdot 1.0 + 2 \cdot 11.1 \cdot 26.4411 \cdot 1.0 \cdot (1 - 0.86))$
 $= 52.907 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 196.875(1.0 \cdot 30.4 - 1.0 \cdot 26.441) - 2 \cdot 11.1$
 $(1.0 \cdot 30.4 - 1.0 \cdot 26.441) \cdot (1 - 0.855)$
 $= 7.667 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 59.75) \cdot (11.1 - 1.25) \cdot 0.8551$

$$= 10.067 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (393.75 - 219.075) \cdot 32.0 \cdot 1.$
 $= 55.895 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.8476 \text{ mm.}$
Wall Thickness per UG16(b),	$tr16b = 3.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 30.9731 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 30.9731 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 8.7600 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.76, \max(30.9731, 3.1)]$
 $= 8.7600 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.8476, 8.76)$
 $= 8.7600 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 265.2 N./mm ²	Passed
Occasional	: 5.7,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 16.0,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $tr = 1.248$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.131$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 1.248$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.131$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 1.248$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.131$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 32.0, tr = 26.441, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.87$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 11.113, tr = 1.248, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.131$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -2 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50/19.60 = 0.689$

Weld Size Calculations, Description: N3-8"

Intermediate Calc. for nozzle/shell Welds Tmin 11.1000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	21.2100 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (52.9067 - 7.6667 + 2 \cdot 11.1 \cdot 0.8551 \cdot (1.0 \cdot 30.4 - 26.4411)) \cdot 138)$
 $= 64718.34 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (10.0671 + 55.895 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 94023.98 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (10.0671 + 0.0 + 0.8551 + (5.7706)) \cdot 138$
 $= 23489.83 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (10.0671 + 0.0 + 0.8551 + 55.895 + (5.7706)) \cdot 138$
 $= 102144.27 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:



$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 550.0 \cdot 30.0 \cdot 0.49 \cdot 138$$

$$= 178707. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 103.9875) \cdot (12.7 - 1.6) \cdot 0.7 \cdot 118$$

$$= 30542. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 32.0 \cdot 0.74 \cdot 138$$

$$= 114667. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 108935. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (178707 + 30542) = 209249 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20289 + 114667 + 108935 + 0) = 243891 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (178707 + 108935 + 0) = 287643 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 209249 Kg, must exceed W = 64718 Kg or W1 = 94023 Kg
 Path 2-2 = 243890 Kg, must exceed W = 64718 Kg or W2 = 23489 Kg
 Path 3-3 = 287642 Kg, must exceed W = 64718 Kg or W3 = 102144 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.1057 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 483.1057 mm.

Input Echo, WRC107/537 Item 1, Description: N3-8" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.6000 mm.
Vessel Diameter	Dv	10800.000 mm.
Vessel Thickness	Tv	32.000 mm.
Design Temperature	T1	82.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.6000 mm.
Nozzle Diameter	Dn	219.075 mm.

Nozzle Thickness	Tn	12.700	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	32.000	mm.
Diameter of Reinforcing Pad	Dpad	550.000	mm.
Design Internal Pressure	Dp	13.500	bars
Include Pressure Thrust		No	

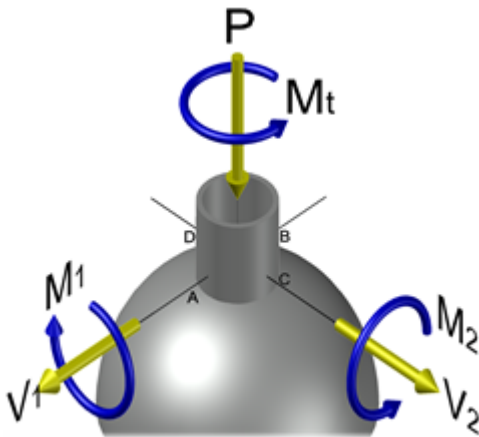
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	1529.1	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1019.4	Kgf
Circumferential Shear	(SUS)	(Vc) V2	1019.4	Kgf
Circumferential Moment	(SUS)	(Mc) M1	570.9	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	570.9	Kg-m.
Torsional Moment	(SUS)	Mt	815.6	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(5416.799 (32.0 - 1.6))}$$

$$= 1558.203 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	1529.1	Kgf
Circumferential Shear	(VC)	V2	1019.4	Kgf
Longitudinal Shear	(VL)	V1	1019.4	Kgf
Circumferential Moment	(MC)	M1	570.9	Kg-m.
Longitudinal Moment	(ML)	M2	570.9	Kg-m.
Torsional Moment		MT	815.6	Kg-m.

Dimensionless Param: U = 0.19 TAU = 9.37 RHO = 4.00 (5.62)

Dimensionless Loads for Spherical Shells at Attachment Junction:



Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.08039	(A,B,C,D)
M(x) / P	SP 4	0.03605	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.28430	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.14238	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.28430	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.14238	(A,B,C,D)
N(y) * T / P	SP 4	0.35086	(A,B,C,D)
M(y) / P	SP 4	0.20645	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.12132	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	0.94683	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.12132	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	0.94683	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Rad. Bend. P		-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-2.1	2.1	2.1	-2.1
Rad. Memb. ML		-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-2.1	2.1	2.1	-2.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-4.0	1.9	1.7	-0.9	-4.0	1.9	1.7	-0.9
Tang. Memb. P		-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4
Tang. Bend. P		-4.8	4.8	-4.8	4.8	-4.8	4.8	-4.8	4.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-14.0	14.0	14.0	-14.0
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-14.0	14.0	14.0	-14.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-20.5	17.1	8.2	-10.3	-20.5	17.1	8.2	-10.3
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tot. Shear		2.2	2.2	1.2	1.2	1.2	1.2	2.2	2.2
Str. Int.		20.7	17.5	8.4	10.5	20.5	17.2	8.9	10.8

Unitless Prm: U = 0.68 TAU = 0.00 (24.27) RHO = 0.00 (2.74)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.10268	(A,B,C,D)
M(x) / P	SR 2	0.05791	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.11739	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.15272	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.11739	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.15272	(A,B,C,D)
N(y) * T / P	SR 2	0.03073	(A,B,C,D)
M(y) / P	SR 2	0.01789	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.03546	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.04510	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.03546	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.04510	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Rad. Bend. P		-5.6	5.6	-5.6	5.6	-5.6	5.6	-5.6	5.6
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-13.7	13.7	13.7	-13.7
Rad. Memb. ML		-1.8	-1.8	1.8	1.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-13.7	13.7	13.7	-13.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-22.7	15.9	8.1	-8.0	-22.7	15.9	8.1	-8.0
Tang. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Tang. Bend. P		-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-4.0	4.0	4.0	-4.0
Tang. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-4.0	4.0	4.0	-4.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-6.8	4.8	2.3	-2.3	-6.8	4.8	2.3	-2.3
Shear VC		0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tot. Shear		0.9	0.9	0.2	0.2	0.2	0.2	0.9	0.9
Str. Int.		22.8	16.0	8.1	8.0	22.7	15.9	8.3	8.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Rad. Pl (SUS)		-1.0	-1.0	0.4	0.4	-1.0	-1.0	0.4	0.4
Rad. Q (SUS)		-2.9	2.9	1.3	-1.3	-2.9	2.9	1.3	-1.3
Long. Pm (SUS)		58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Long. Pl (SUS)		-1.7	-1.7	-1.1	-1.1	-1.7	-1.7	-1.1	-1.1
Long. Q (SUS)		-18.8	18.8	9.3	-9.3	-18.8	18.8	9.3	-9.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Pm (SUS)		58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Pm+Pl (SUS)		57.7	57.7	59.0	59.0	57.7	57.7	59.0	59.0
Pm+Pl+Q (Total)		54.8	75.9	66.9	57.7	54.6	75.7	67.3	58.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.44	138.00	Passed
Pm+Pl (SUS)	58.97	207.00	Passed
Pm+Pl+Q (TOTAL)	75.89	414.00	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Rad.	Pl (SUS)	-3.4	-3.4	0.1	0.1	-3.4	-3.4	0.1	0.1
Rad.	Q (SUS)	-19.3	19.3	8.0	-8.0	-19.3	19.3	8.0	-8.0
Long.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Long.	Pl (SUS)	-1.0	-1.0	0.0	0.0	-1.0	-1.0	0.0	0.0
Long.	Q (SUS)	-5.8	5.8	2.3	-2.3	-5.8	5.8	2.3	-2.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Pm (SUS)		119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Pm+Pl (SUS)		119.0	119.0	120.4	120.4	119.0	119.0	120.4	120.4
Pm+Pl+Q (Total)		113.2	135.9	128.1	117.7	113.1	135.8	128.2	117.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.95	138.00	Passed
Pm+Pl (SUS)	120.39	207.00	Passed
Pm+Pl+Q (TOTAL)	135.92	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-6"

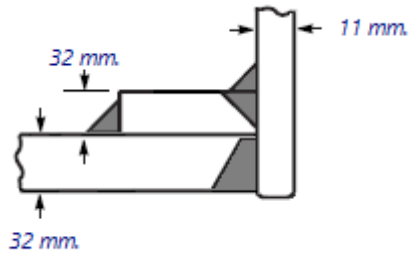
From: 130

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1250.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	550.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	350.0000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	30.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		90.8625	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$
 $= 26.4411 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 13.5 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 13.5)$
 $= 0.9583 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	306.8141 mm.
Parallel to Vessel Wall, opening length	d	153.4070 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		55.4320 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.)$
 $= 0.855$

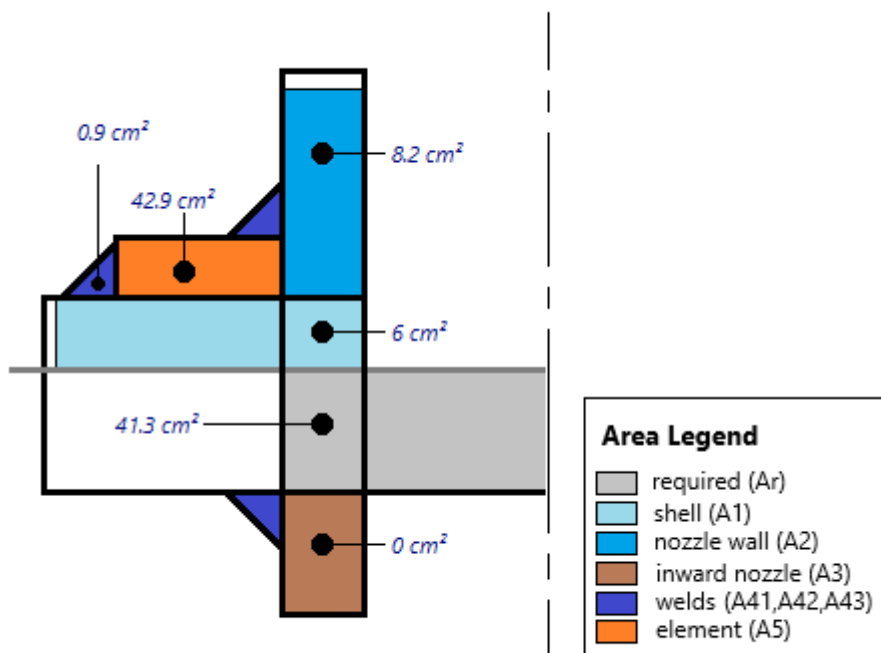


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	41.281	NA	NA
Area in Shell	A1	5.966	NA	NA
Area in Nozzle Wall	A2	8.151	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	42.935	NA	NA
TOTAL AREA AVAILABLE	Atot	57.907	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 77.09 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 250.4928 32.0000 mm.
 Based on given Pad Diameter: 350.0000 19.6084 mm.
 Based on the Estimated Diameter Limit: 305.2265 19.8432 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (153.407 \cdot 26.4411 \cdot 1.0 + 2 \cdot 9.3728 \cdot 26.4411 \cdot 1.0 \cdot (1 - 0.86))$
 $= 41.281 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 153.407(1.0 \cdot 30.4 - 1.0 \cdot 26.441) - 2 \cdot 9.373$
 $(1.0 \cdot 30.4 - 1.0 \cdot 26.441) \cdot (1 - 0.855)$
 $= 5.966 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 55.43) \cdot (9.37 - 0.96) \cdot 0.8551 / \sin(78.1)$

= 8.151 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.$
 = 0.855 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(306.8141 - 172.6387) \cdot 32.0 \cdot 1.$
 = 42.935 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.958, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.958, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.958, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 32.0, tr = 26.441, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.87$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.601, tr = 0.958, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -2 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $13.50 / 19.60 = 0.689$

Weld Size Calculations, Description: N4-6"

Intermediate Calc. for nozzle/shell Welds Tmin 9.3728 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	21.2100 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (41.2809 - 5.9657 + 2 \cdot 9.3728 \cdot 0.8551 \cdot (1.0 \cdot 30.4 - 26.4411)) \cdot 138)$
 $= 50587.85 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (8.1509 + 42.9353 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 73090.90 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (8.1509 + 0.0 + 0.8551 + (4.8727)) \cdot 138$
 $= 19529.82 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (8.1509 + 0.0 + 0.8551 + 42.9353 + (4.8727)) \cdot 138$
 $= 79947.65 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 172.6387 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15988. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 350.0 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 113723. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 81.5114) \cdot (10.9728 - 1.6) \cdot 0.7 \cdot 118$
 $= 20215. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 172.6387 \cdot 32.0 \cdot 0.74 \cdot 138$
 $= 90361. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 172.6387 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 85845. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (113723 + 20215) = 133938 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (15988 + 90361 + 85845 + 0) = 192194 \text{ Kgf}$

Failure Path [Path 3-3]:
 = (Spew + Tngw + Sinw)
 = (113723 + 85845 + 0) = 199568 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 133938 Kgf, must exceed W = 50587 Kgf or W1 = 73090 Kgf
 Path 2-2 = 192194 Kgf, must exceed W = 50587 Kgf or W2 = 19529 Kgf
 Path 3-3 = 199567 Kgf, must exceed W = 50587 Kgf or W3 = 79947 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 = $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 = $13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 13.5)$
 = 26.4411 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $13.5 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 13.5)$
 = 0.9583 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	299.0588 mm.
Parallel to Vessel Wall, opening length	d	149.5294 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		55.4320 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

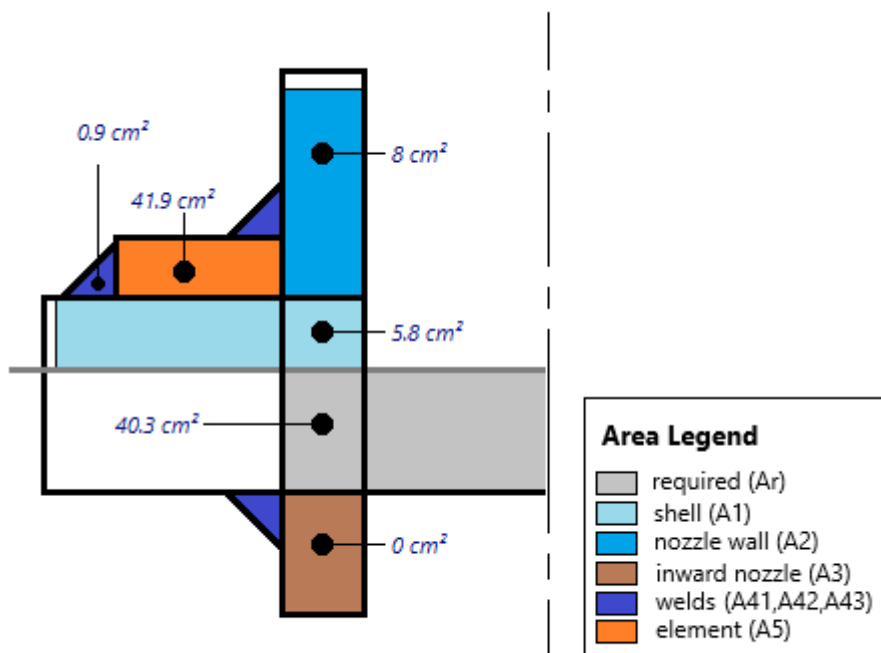


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	40.256	NA	NA
Area in Shell	A1	5.812	NA	NA
Area in Nozzle Wall	A2	7.977	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	41.850	NA	NA
TOTAL AREA AVAILABLE	Atot	56.494	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 248.3134 32.0000 mm.
 Based on given Pad Diameter: 350.0000 19.5837 mm.
 Based on the Estimated Diameter Limit: 297.4713 19.8243 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (149.5294 \cdot 26.4411 \cdot 1.0 + 2 \cdot 9.3728 \cdot 26.4411 \cdot 1.0 \cdot (1 - 0.86))$
 $= 40.256 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 149.529(1.0 \cdot 30.4 - 1.0 \cdot 26.441) - 2 \cdot 9.373$
 $(1.0 \cdot 30.4 - 1.0 \cdot 26.441) \cdot (1 - 0.855)$
 $= 5.812 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 55.43) \cdot (9.37 - 0.96) \cdot 0.8551$

$$= 7.977 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (299.0588 - 168.275) \cdot 32.0 \cdot 1.$
 $= 41.850 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.5583 \text{ mm}.$
Wall Thickness per UG16(b),	$tr16b = 3.1000 \text{ mm}.$
Wall Thickness, shell/head, internal pressure	$trb1 = 30.9731 \text{ mm}.$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 30.9731 \text{ mm}.$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
Wall Thickness per table UG-45	$tb3 = 7.8200 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.82, \max(30.9731, 3.1)]$
 $= 7.8200 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.5583, 7.82)$
 $= 7.8200 \text{ mm}.$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm}.$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 32.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 262.2 N./mm ²	Passed
Occasional	: 5.1,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 18.5,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N4-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.3728 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}.$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$21.2100 = 0.7 \cdot W_p \text{ mm}.$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (40.2556 - 5.8122 + 2 \cdot 9.3728 \cdot 0.8551 \cdot (1.0 \cdot 30.4 - 26.4411)) \cdot 138)$
 $= 49361.11 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot Sv$
 $= (7.9766 + 41.8501 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 71318.35 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$

$$= (7.9766 + 0.0 + 0.8551 + (4.8727)) \cdot 138$$

$$= 19284.43 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (7.9766 + 0.0 + 0.8551 + 41.8501 + (4.8727)) \cdot 138$$

$$= 78175.09 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 350.0 \cdot 30.0 \cdot 0.49 \cdot 138$$

$$= 113723. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 79.4511) \cdot (10.9728 - 1.6) \cdot 0.7 \cdot 118$$

$$= 19704. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 32.0 \cdot 0.74 \cdot 138$$

$$= 88077. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 83675. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (113723 + 19704) = 133427 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 88077 + 83675 + 0) = 187336 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (113723 + 83675 + 0) = 197398 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 133427 Kgf, must exceed W = 49361 Kgf	or W1 = 71318 Kgf
Path 2-2 = 187336 Kgf, must exceed W = 49361 Kgf	or W2 = 19284 Kgf
Path 3-3 = 197397 Kgf, must exceed W = 49361 Kgf	or W3 = 78175 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 20.6309 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 603.5103 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N4-6" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	10800.000	mm.
Vessel Thickness	Tv	32.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	168.275	mm.
Nozzle Thickness	Tn	10.973	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	32.000	mm.
Diameter of Reinforcing Pad	Dpad	350.000	mm.
Design Internal Pressure	Dp	13.500	bars
Include Pressure Thrust		No	

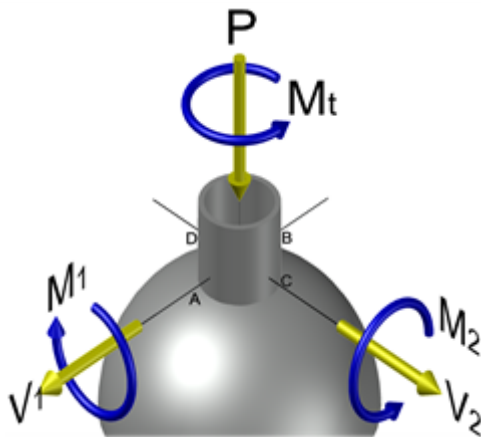
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	1146.8	Kgf
Longitudinal Shear (SUS)	(V1) V1	764.5	Kgf
Circumferential Shear (SUS)	(Vc) V2	764.5	Kgf
Circumferential Moment (SUS)	(Mc) M1	321.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	321.1	Kg-m.
Torsional Moment (SUS)	Mt	458.7	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(5416.799 (32.0 - 1.6))}$$

$$= 1507.403 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	1146.8	Kgf
Circumferential Shear	(VC) V2	764.5	Kgf
Longitudinal Shear	(VL) V1	764.5	Kgf
Circumferential Moment	(MC) M1	321.1	Kg-m.
Longitudinal Moment	(ML) M2	321.1	Kg-m.
Torsional Moment	MT	458.7	Kg-m.

Dimensionless Param: U = 0.14 TAU = 8.48 RHO = 4.00 (6.66)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.09133	(A,B,C,D)
$M(x) / P$	SP 4	0.04055	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.36480	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.18171	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.36480	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.18171	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34589	(A,B,C,D)
$M(y) / P$	SP 4	0.25131	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10206	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.26147	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10206	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.26147	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Rad. Bend.	P	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-1.5	1.5	1.5	-1.5
Rad. Memb.	ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-1.5	1.5	1.5	-1.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-3.0	1.4	1.1	-0.6	-3.0	1.4	1.1	-0.6



Tang. Memb. P	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
Tang. Bend. P	-4.4	4.4	-4.4	4.4	-4.4	4.4	-4.4	4.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-10.5	10.5	10.5	-10.5
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-10.5	10.5	10.5	-10.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-16.0	13.7	5.3	-7.0	-16.0	13.7	5.3	-7.0
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tot. Shear	2.1	2.1	1.2	1.2	1.2	1.2	2.1	2.1
Str. Int.	16.3	14.1	5.6	7.2	16.1	13.8	6.1	7.7

Unitless Prm: U = 0.43 TAU = 0.00 (18.17) RHO = 0.00 (3.24)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.15097	(A,B,C,D)
M(x) / P	SR 2	0.09809	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15254	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.30175	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15254	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.30175	(A,B,C,D)
N(y) * T / P	SR 2	0.04542	(A,B,C,D)
M(y) / P	SR 2	0.02955	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04633	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.08980	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04633	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.08980	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8
Rad. Bend. P		-7.2	7.2	-7.2	7.2	-7.2	7.2	-7.2	7.2
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Rad. Bend. MC		0.0	0.0	0.0	0.0	-15.2	15.2	15.2	-15.2
Rad. Memb. ML		-1.3	-1.3	1.3	1.3	0.0	0.0	0.0	0.0
Rad. Bend. ML		-15.2	15.2	15.2	-15.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-25.5	19.2	7.5	-8.6	-25.5	19.2	7.5	-8.6
Tang. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Tang. Bend. P		-2.2	2.2	-2.2	2.2	-2.2	2.2	-2.2	2.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Tang. Bend. MC		0.0	0.0	0.0	0.0	-4.5	4.5	4.5	-4.5
Tang. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-4.5	4.5	4.5	-4.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-7.6	5.7	2.2	-2.5	-7.6	5.7	2.2	-2.5
Shear VC		0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tot. Shear		1.2	1.2	0.3	0.3	0.3	0.3	1.2	1.2

Str. Int.	25.6	19.4	7.5	8.6	25.5	19.3	7.8	8.8
-----------	------	------	-----	-----	------	------	-----	-----

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Rad.	Pl (SUS)	-0.8	-0.8	0.2	0.2	-0.8	-0.8	0.2	0.2
Rad.	Q (SUS)	-2.2	2.2	0.8	-0.8	-2.2	2.2	0.8	-0.8
Long.	Pm (SUS)	58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Long.	Pl (SUS)	-1.1	-1.1	-0.9	-0.9	-1.1	-1.1	-0.9	-0.9
Long.	Q (SUS)	-14.9	14.9	6.2	-6.2	-14.9	14.9	6.2	-6.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		58.4	58.4	58.4	58.4	58.4	58.4	58.4	58.4
Pm+Pl (SUS)		58.0	58.0	58.8	58.8	58.0	58.0	58.8	58.8
Pm+Pl+Q (Total)		55.8	72.5	64.0	58.1	55.6	72.3	64.6	58.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.44	138.00	Passed
Pm+Pl (SUS)	58.85	207.00	Passed
Pm+Pl+Q (TOTAL)	72.51	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Rad.	Pl (SUS)	-3.1	-3.1	-0.6	-0.6	-3.1	-3.1	-0.6	-0.6
Rad.	Q (SUS)	-22.4	22.4	8.0	-8.0	-22.4	22.4	8.0	-8.0
Long.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Long.	Pl (SUS)	-0.9	-0.9	-0.2	-0.2	-0.9	-0.9	-0.2	-0.2
Long.	Q (SUS)	-6.7	6.7	2.4	-2.4	-6.7	6.7	2.4	-2.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Pm (SUS)		119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Pm+Pl (SUS)		119.1	119.1	120.1	120.1	119.1	119.1	120.1	120.1
Pm+Pl+Q (Total)		112.4	139.3	127.5	117.4	112.3	139.2	127.7	117.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.95	138.00	Passed
Pm+Pl (SUS)	120.08	207.00	Passed
Pm+Pl+Q (TOTAL)	139.30	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N5-2"

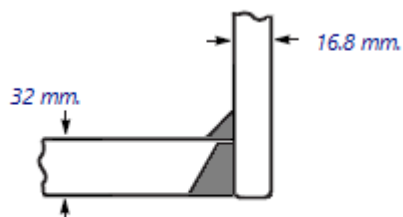
From: 130

Pressure for Reinforcement Calculations	P	13.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	750.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$
 $= 26.4411$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2654$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 147.3192 mm.
 Parallel to Vessel Wall Rn+tn+t 73.6596 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 37.9100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

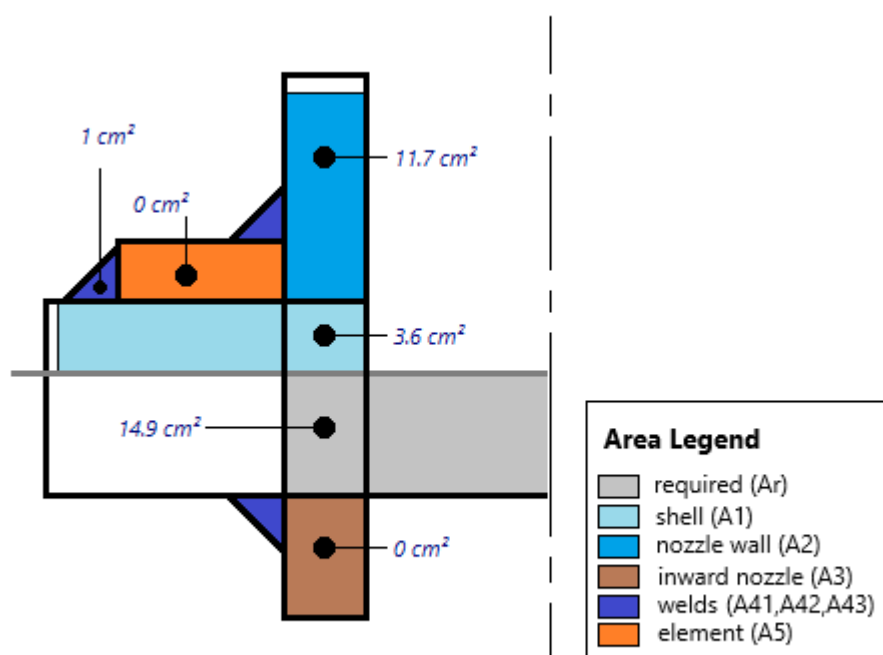


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	14.858	NA	NA



Area in Shell	A1	3.608	NA	NA
Area in Nozzle Wall	A2	11.724	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.332	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 73.95 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (56.1912 \cdot 26.4411 \cdot 1.0 + 2 \cdot 15.164 \cdot 26.4411 \cdot 1.0 \cdot (1 - 1.))$
 $= 14.858 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 91.128(1.0 \cdot 30.4 - 1.0 \cdot 26.4411) - 2 \cdot 15.164$
 $(1.0 \cdot 30.4 - 1.0 \cdot 26.4411) \cdot (1 - 1.)$
 $= 3.608 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 37.91)(15.16 - 0.27)1/\sin(74.5)$
 $= 11.724 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.764, tr = 0.265, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.018, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 13.50/51.00 = 0.265

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.1640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (14.8577 - 3.6077 + 2 \cdot 15.164 \cdot 1. \cdot \\
 &\quad (1.0 \cdot 30.4 - 26.4411)) \cdot 138) \\
 &= 17520.36 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (11.724 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138 \\
 &= 17904.93 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (11.724 + 0.0 + 1. + (9.2195)) \cdot 138 \\
 &= 30878.51 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (11.724 + 0.0 + 1. + 0.0 + (9.2195)) \cdot 138 \\
 &= 30878.51 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 87.7498 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 9504. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 35.9852) \cdot (16.764 - 1.6) \cdot 0.7 \cdot 138 \\
 &= 16886. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 87.7498 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138 \\
 &= 43634. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9504 + 16886) = 26390 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (9504 + 0 + 43634 + 0) = 53138 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (9504 + 43634 + 0) = 53138 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 26390	Kgf, must exceed W = 17520 Kgf	or W1 = 17904 Kgf
Path 2-2 = 53137	Kgf, must exceed W = 17520 Kgf	or W2 = 30878 Kgf
Path 3-3 = 53137	Kgf, must exceed W = 17520 Kgf	or W3 = 30878 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 13.5 \cdot 6003.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 13.5)$
 $= 26.4411 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 (\exp([13.5 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2654 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 145.1280 mm.
 Parallel to Vessel Wall Rn+tn+t 72.5640 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 37.9100 mm.

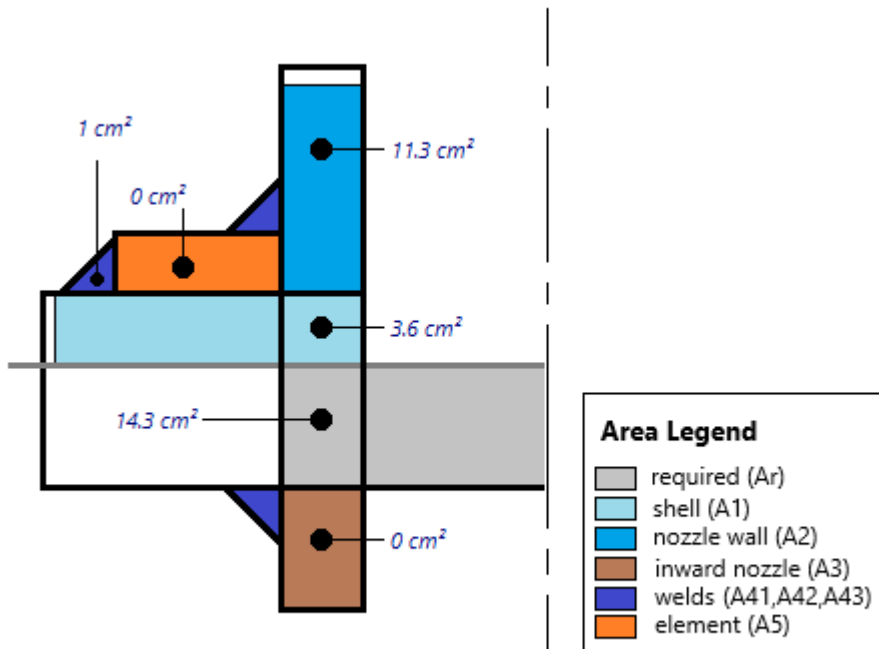


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	14.278	NA	NA
Area in Shell	A1	3.608	NA	NA
Area in Nozzle Wall	A2	11.296	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.904	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (54.0 \cdot 26.4411 \cdot 1.0 + 2 \cdot 15.164 \cdot 26.4411 \cdot 1.0 \cdot (1 - 1.))$$

$$= 14.278 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 91.128(1.0 \cdot 30.4 - 1.0 \cdot 26.441) - 2 \cdot 15.164$$

$$(1.0 \cdot 30.4 - 1.0 \cdot 26.441) \cdot (1 - 1.)$$

$$= 3.608 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.91)(15.16 - 0.27) \cdot 1.$$

$$= 11.296 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.8654 \text{ mm}.$
 Wall Thickness per UG16(b), $tr16b = 3.1000 \text{ mm}.$
 Wall Thickness, shell/head, internal pressure $trb1 = 30.9731 \text{ mm}.$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 30.9731 \text{ mm}.$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.1000 \text{ mm}.$
 Wall Thickness per table UG-45 $tb3 = 6.4000 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.4, \max(30.9731, 3.1)]$$

$$= 6.4000 \text{ mm}.$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.8654, 6.4)$$

$$= 6.4000 \text{ mm}.$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 12.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 332.8 N./mm ²	Passed
Occasional	: 0.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 9.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds $Tmin = 15.1640 \text{ mm}.$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot Wo \text{ mm}.$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (14.2783 - 3.6077 + 2 \cdot 15.164 \cdot 1. \cdot$$

$$(1.0 \cdot 30.4 - 26.4411)) 138)$$

$$= 16705.09 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (11.2959 + 0.0 + 1. - 0.0 \cdot 1.) \cdot 138$$

$$= 17302.52 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (11.2959 + 0.0 + 1. + (9.2195)) \cdot 138$$

$$= 30276.10 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (11.2959 + 0.0 + 1. + 0.0 + (9.2195)) \cdot 138$$

$$= 30276.10 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.582) \cdot (16.764 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16228. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.328 \cdot (32.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 41932. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9133 + 16228) = 25361 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9133 + 0 + 41932 + 0) = 51066 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9133 + 41932 + 0) = 51066 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25360	Kgf, must exceed W = 16705 Kgf	or W1 = 17302 Kgf
Path 2-2 = 51065	Kgf, must exceed W = 16705 Kgf	or W2 = 30276 Kgf
Path 3-3 = 51065	Kgf, must exceed W = 16705 Kgf	or W3 = 30276 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.3148 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 795.6046 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N5-2" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.6000 mm.

Vessel Diameter	Dv	10800.000	mm.
Vessel Thickness	Tv	32.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.6000 mm.
Nozzle Diameter	Dn	50.800 mm.
Nozzle Thickness	Tn	16.764 mm.
Nozzle Material		SA-105
Nozzle UNS Number		K03504
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²
Design Internal Pressure	Dp	13.500 bars
Include Pressure Thrust		No

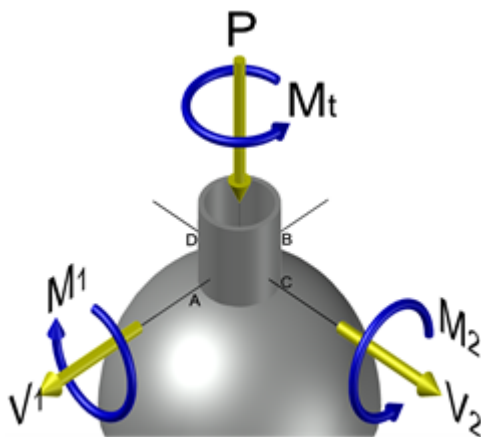
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.3	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.8	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.8	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1
 Warning - The ratio Ro/Rm, 0.0078, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword

and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(5416.799 (32.0 - 1.6))}$$

$$= 1423.456 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	382.3	Kgf
Circumferential Shear	(VC) V2	254.8	Kgf
Longitudinal Shear	(VL) V1	254.8	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Param: U = 0.10 TAU = 5.00 (2.28) RHO = 2.00

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 3	0.19453	(A,B,C,D)
M(x) / P	SP 3	0.15983	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 3	0.61775	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 3	0.60839	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 3	0.61775	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 3	0.60839	(A,B,C,D)
N(y) * T / P	SP 3	0.19167	(A,B,C,D)
M(y) / P	SP 3	0.15975	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 3	0.08788	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 3	1.15835	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 3	0.08788	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 3	1.15835	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8
Rad.	Bend. P	-3.9	3.9	-3.9	3.9	-3.9	3.9	-3.9	3.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-3.4	3.4	3.4	-3.4
Rad.	Memb. ML	-0.6	-0.6	0.6	0.6	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-3.4	3.4	3.4	-3.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-8.7	5.9	-0.7	0.3	-8.7	5.9	-0.7	0.3
Tang.	Memb. P	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8
Tang.	Bend. P	-3.9	3.9	-3.9	3.9	-3.9	3.9	-3.9	3.9
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-6.5	6.5	6.5	-6.5
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-6.5	6.5	6.5	-6.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-11.2	9.5	1.9	-3.3	-11.2	9.5	1.9	-3.3
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear		2.1	2.1	0.9	0.9	0.9	0.9	2.1	2.1
Str. Int.		12.4	10.5	3.1	3.9	11.5	9.7	4.9	5.5



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Rad.	Pl (SUS)	-1.4	-1.4	-0.2	-0.2	-1.4	-1.4	-0.2	-0.2
Rad.	Q (SUS)	-7.3	7.3	-0.5	0.5	-7.3	7.3	-0.5	0.5
Long.	Pm (SUS)	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Long.	Pl (SUS)	-0.9	-0.9	-0.7	-0.7	-0.9	-0.9	-0.7	-0.7
Long.	Q (SUS)	-10.4	10.4	2.6	-2.6	-10.4	10.4	2.6	-2.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
Pm+Pl (SUS)		119.5	119.5	120.2	120.2	119.5	119.5	120.2	120.2
Pm+Pl+Q (Total)		112.5	130.4	122.1	120.4	111.5	129.7	123.0	121.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.95	138.00	Passed
Pm+Pl (SUS)	120.16	207.00	Passed
Pm+Pl+Q (TOTAL)	130.42	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: